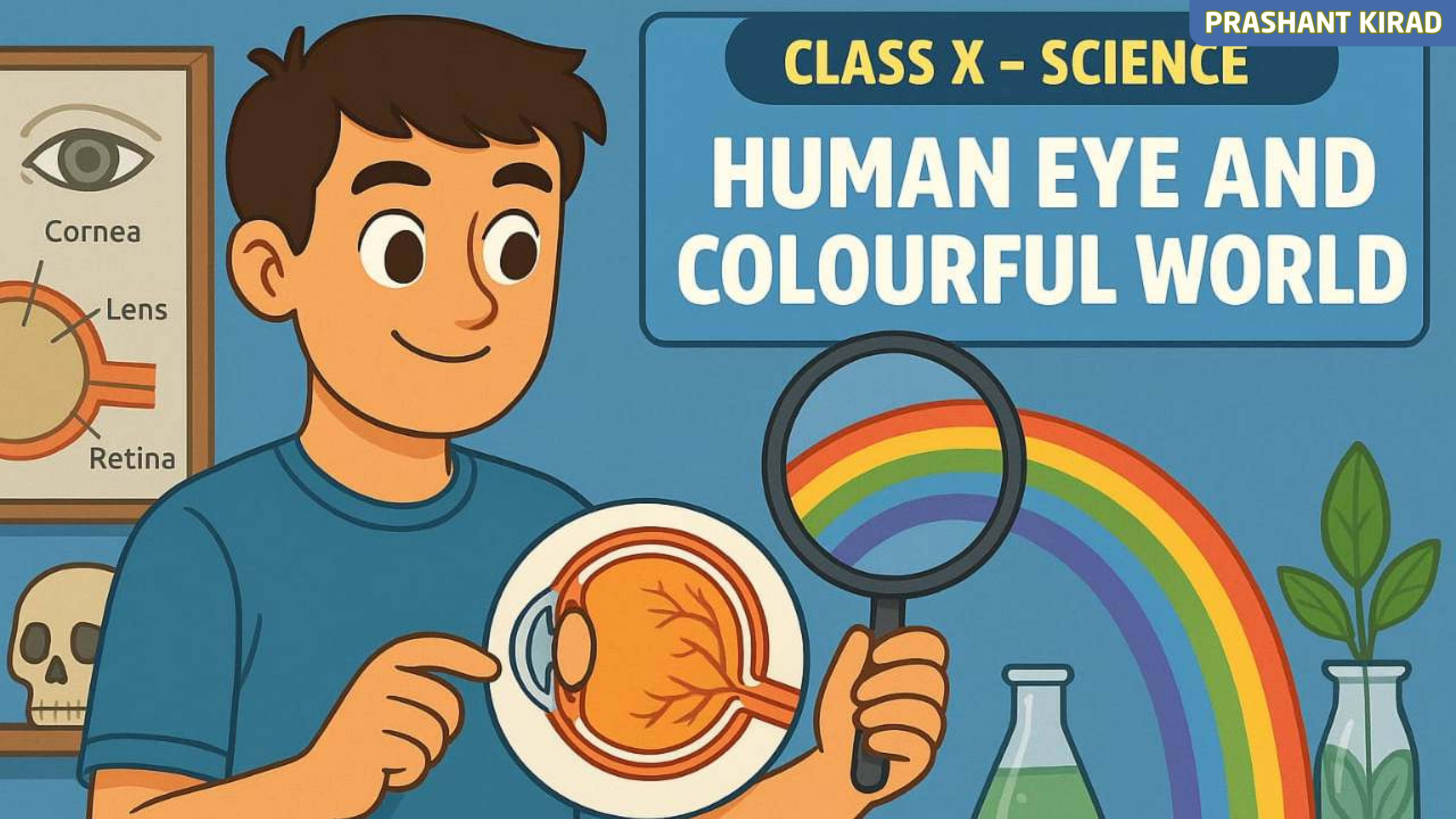
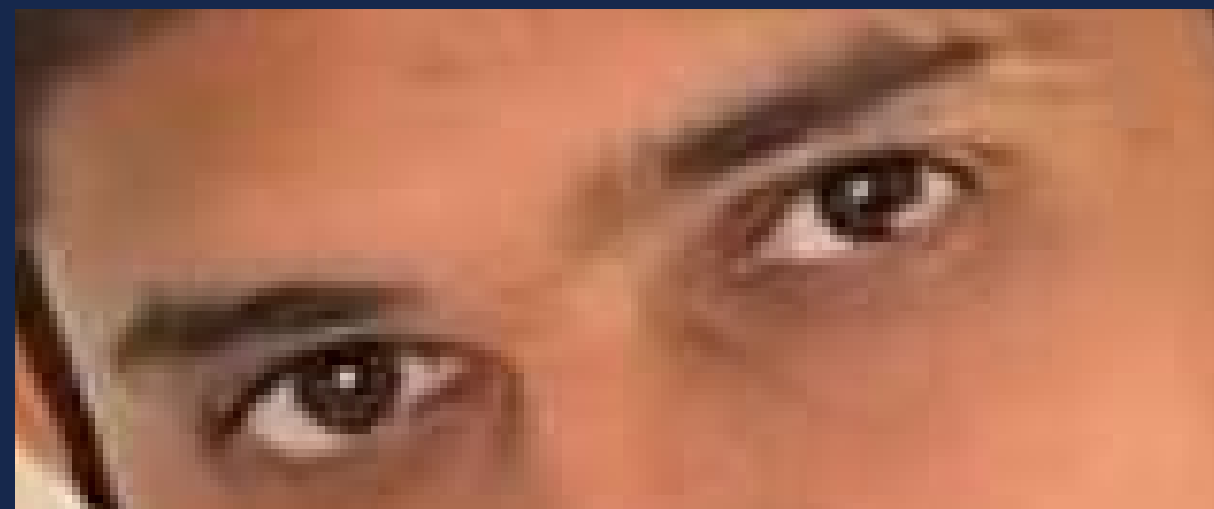
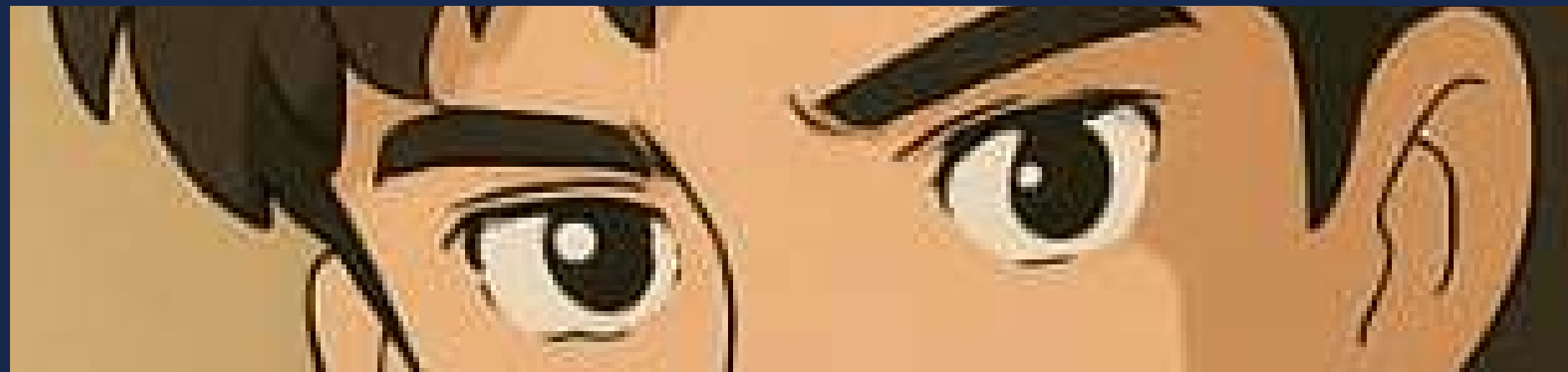


CLASS X – SCIENCE

HUMAN EYE AND COLOURFUL WORLD



WHO'S EYES ARE THESE?

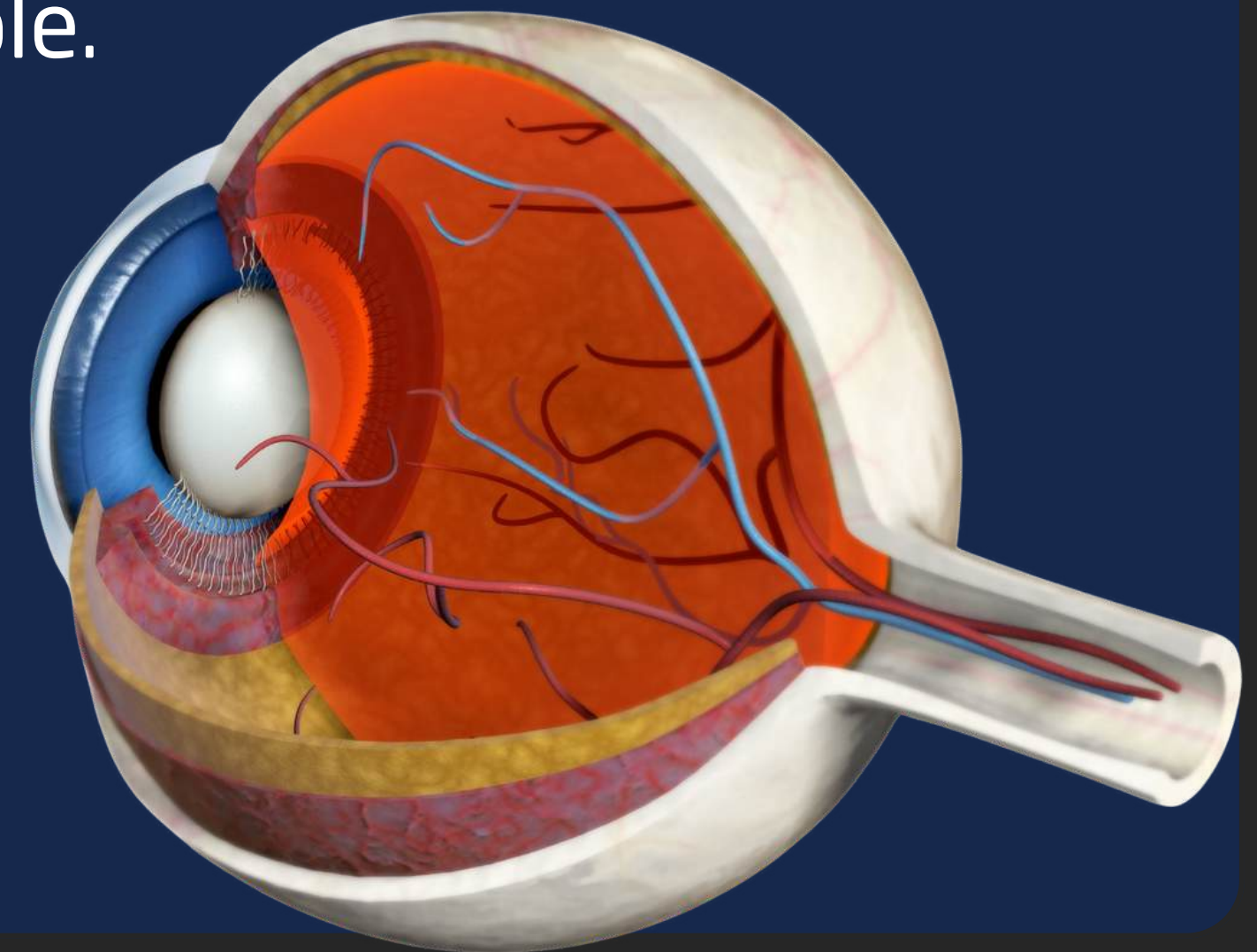


THE HUMAN EYE

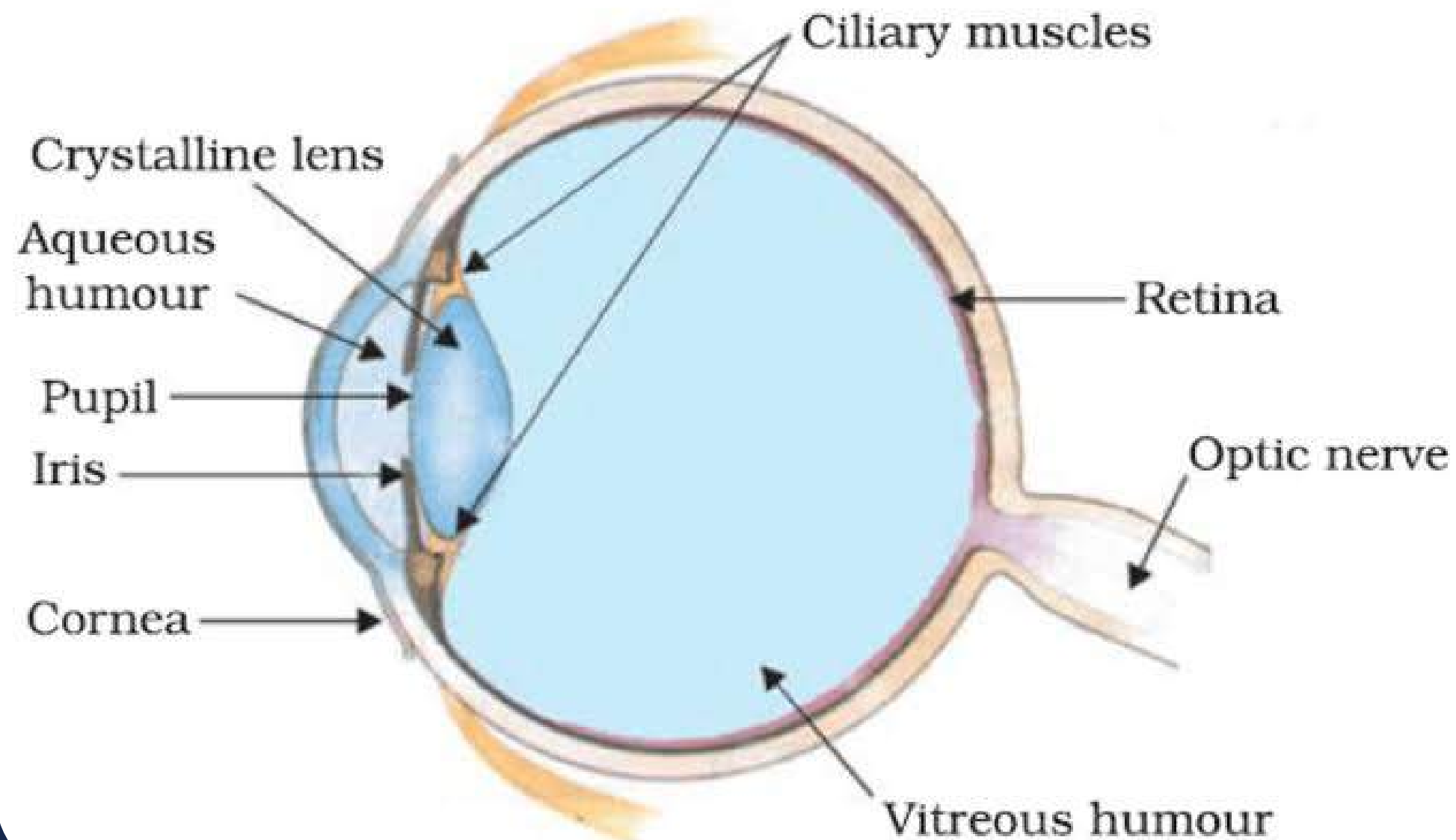
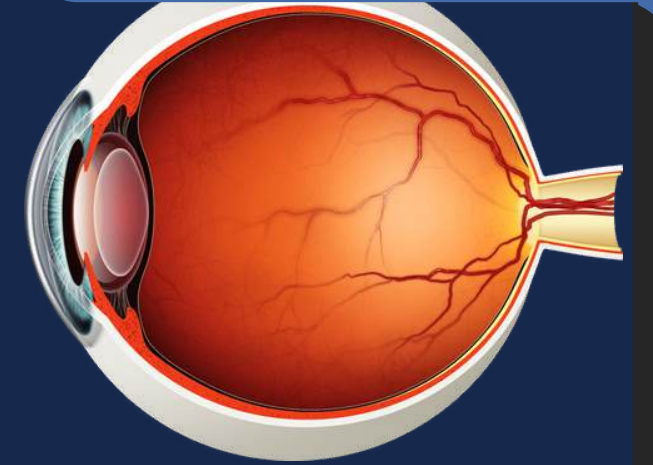
- One of the most sensitive and important sense organs.
- Enables us to see the world and perceive colours.
- Other senses (smell, sound, touch) can help identify objects, but not colour—making vision unique and irreplaceable.

Advantage of eyes in front of the face:

- It gives a wider field of view.
- It provides three-dimensional view



STRUCTURE OF HUMAN EYE



Advantage of eyes in front of the face:

- It gives a wider field of view.
- It provides three-dimensional view



PARTS OF HUMAN EYE

Part of the Eye	Description
Cornea	Transparent, curved membrane at the front; allows light to enter and causes major refraction.
Eyeball	Roughly spherical, about 2.3 cm in diameter.
Iris	Coloured muscular diaphragm behind cornea; controls pupil size and regulates light entry.
Pupil	Opening in the iris; adjusts to control light entering the eye.
Lens	Transparent crystalline lens; fine-tunes focal length to focus image on retina.
Retina	Lens system forms an image on a light-sensitive screen called the retina. Inner light-sensitive layer with rods and cones; converts light to signals.

Part of the Eye	Description
Rods	Detect dim light; responsible for black-and-white vision.
Cones	Detect bright light; responsible for color vision.
Sclera	Tough, white outer covering; protects and maintains the eye's shape.
Optic Nerve	Carries signals from the retina to the brain for image formation.
Ciliary Muscles	Control the lens's shape to adjust focal length for near or distant vision.
Aqueous Humor	Clear fluid between cornea and lens; maintains pressure and refracts light.
Vitreous Humor	Transparent gel between lens and retina; provides shape and supports the retina.
Blind Spot	Area where the optic nerve exits; no photoreceptor cells, so no vision here.

Light enters through the **cornea**.



Major refraction occurs at corneal surface.



Light passes through **aqueous humour, pupil, and lens**.



Lens adjusts focal length to focus image on **retina**.



Inverted real image is formed on retina.



Light-sensitive cells on retina convert light to electrical signals.



Optic nerve carries signals to brain.



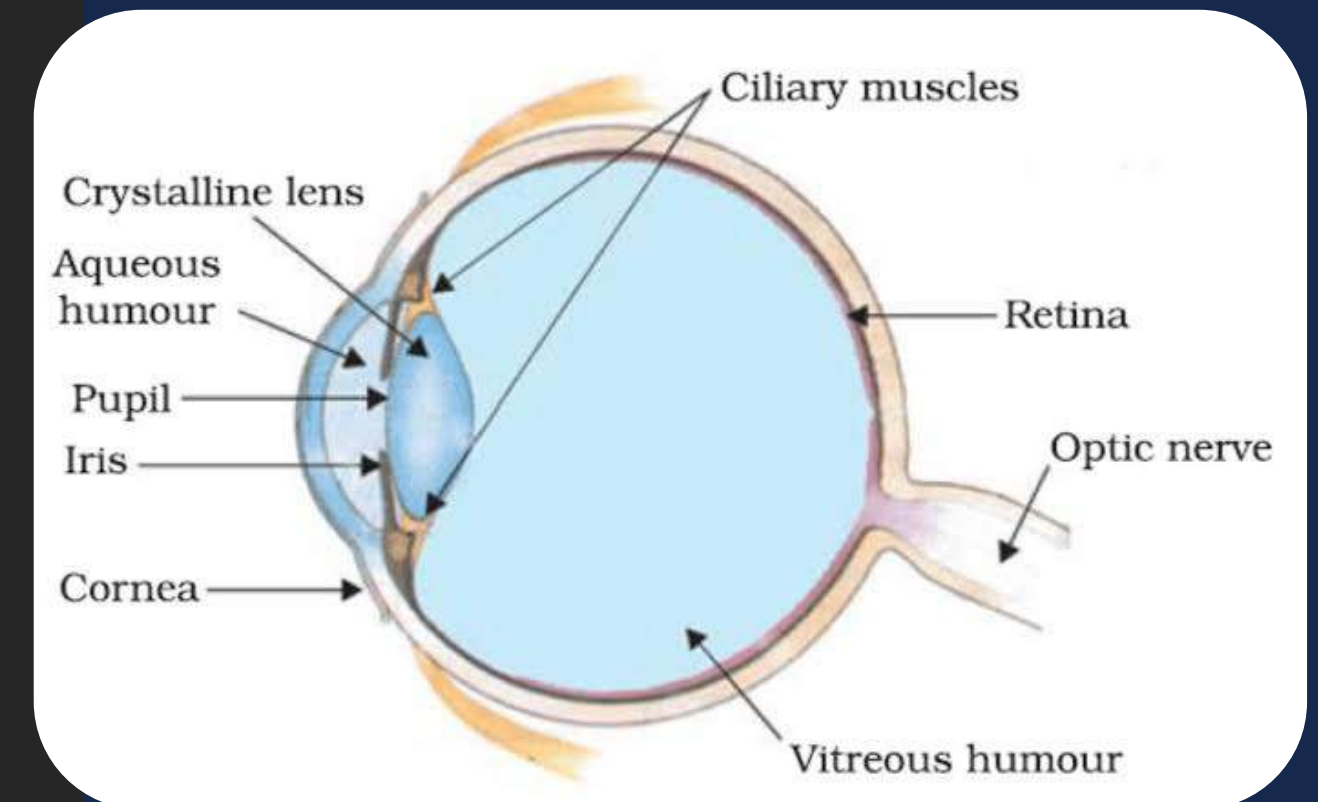
Brain processes these signals → we see the actual upright object.



The image formed on retina is real and inverted.



But our **brain** interprets it as upright and correct.



CBSE 2017, 2020, 2022, 2023

Q. The human eye forms the image of an object at its:

- (a) Iris
- (b) Retina
- (c) Pupil
- (d) Cornea

CBSE 2015, 2018, 2020 (Sample Paper), 2021, 2023

Q. State the functions of:

- (i) Cornea
- (ii) Iris
- (iii) Ciliary muscles
- (iv) Retina

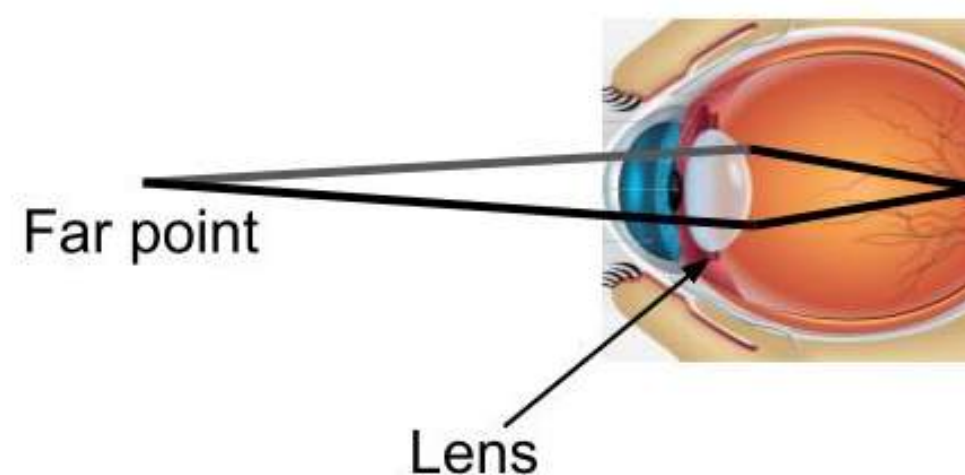
**CBSE 2021 (Term 1), 2022 (Sample Paper), 2023 (Pre-board),
2024 (Sample Paper)**

Q. A person gets out in the sunlight from a dark room. How does his pupil regulate and control the light entering the eye?

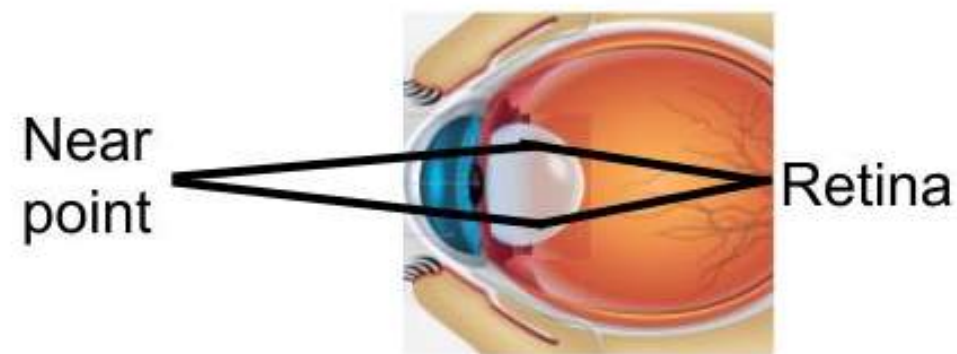
- (a) The size of the pupil will decrease, and less light will enter the eye
- (b) The size of the pupil will decrease, and more light will enter the eye
- (c) The size of the pupil will remain the same, but more light will enter the eye
- (d) The size of the pupil will remain the same, but less light will enter the eye

POWER OF ACCOMMODATION

“Accommodation is the ability of the eye lens to adjust its focal length to focus on near and far objects clearly.”
This ability declines with age.”



- Far Object
- Expanded Eye lens
- Focal length increases and power decreases
- **Ciliary muscles relax** → lens becomes thin → focal length increases

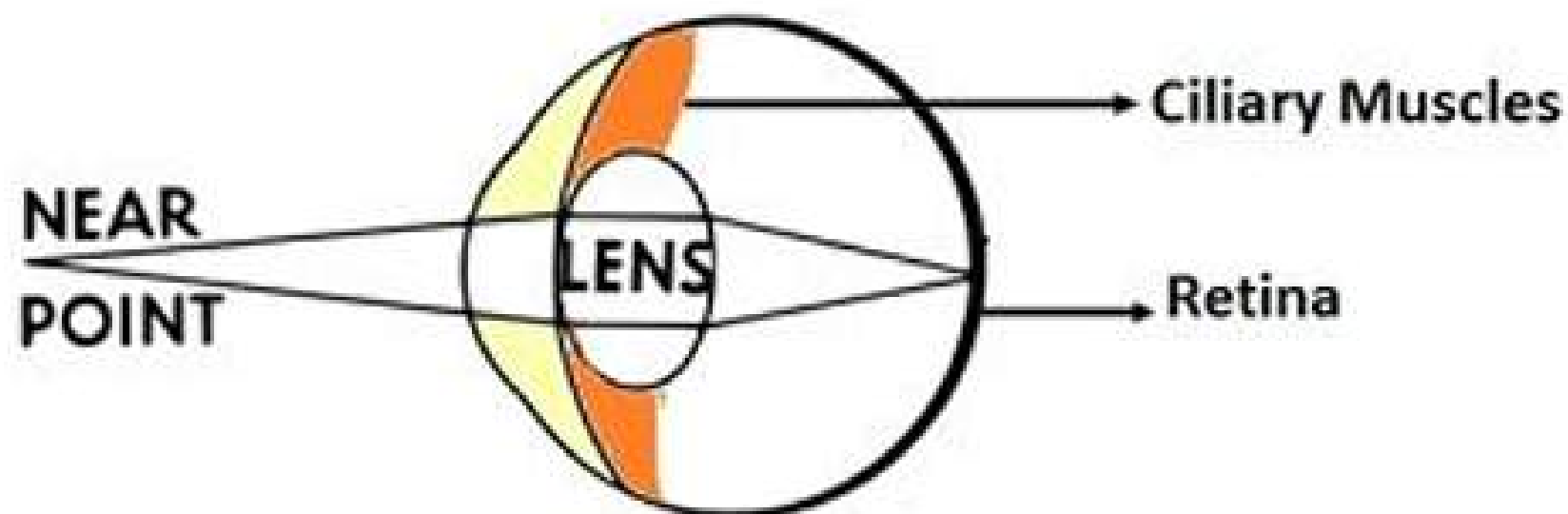


- Nearby Object
- Contracted Eye lens
- Focal length decreases and power increases
- **Ciliary muscles contract** → lens becomes thicker → focal length decreases

- **Lens curvature changes = focal length changes.**
- **Accommodation = focusing mechanism of the eye.**
- **25 cm to ∞ = range of clear vision for a normal eye.**

NEAR POINT

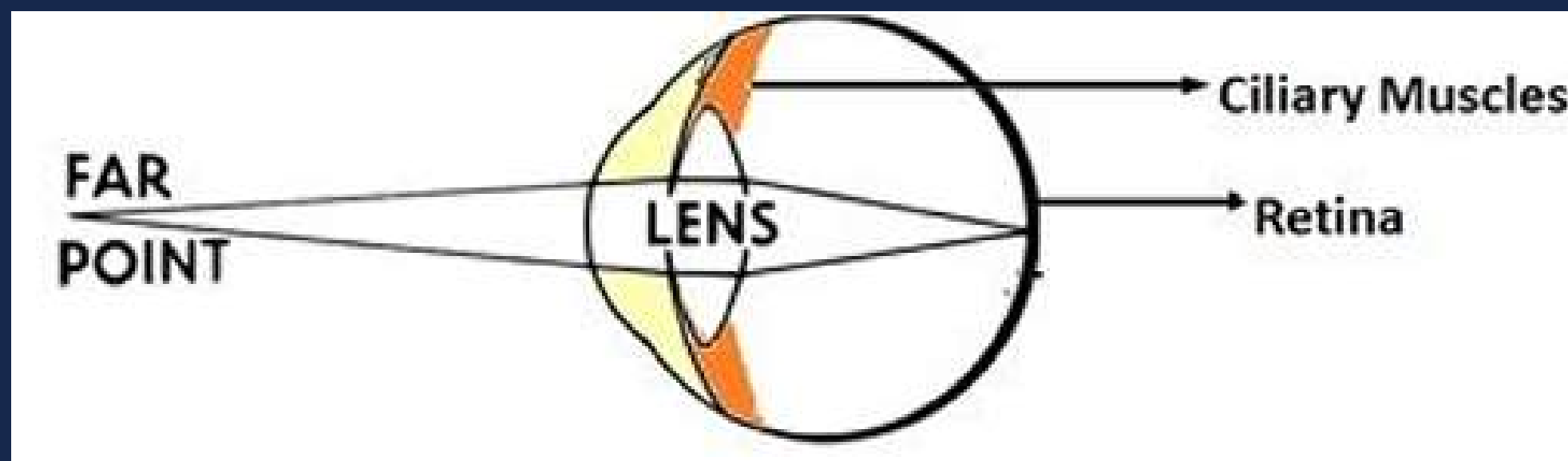
- **The minimum distance at which objects can be seen most distinctly without strain is called the near point of the eye.**
- **It is also called the least distance of distinct vision.**
- *For a normal human eye, the near point is 25 cm.*
- **Below this distance, the image appears blurry and causes eye strain, as the lens cannot decrease its focal length further.**



Object Distance	Ciliary Muscles	Lens Shape	Focal Length	Result
Near Object	Contracted	Thicker	Decreases	Nearby object seen clearly

FAR POINT

- **The farthest point up to which the eye can see objects clearly is called the far point of the eye.**
- *For a normal eye, the far point is at infinity.*
- **The eye can see all objects clearly between the near point (25 cm) and far point (infinity).**



Object Distance	Ciliary Muscles	Lens Shape	Focal Length	Result
Distant Object	Relaxed	Thinner	Increases	Distant object seen clearly

**CBSE 2012, 2013, 2015 (Pre-board), 2016, 2018, 2020
(Sample Paper), 2021 (Term 1), 2022 (Sample Paper), 2023
(Pre-board), 2024 (Sample Paper)**

Q. The least distance of distinct vision for a normal human eye is:

- A. 10 cm**
- B. 15 cm**
- C. 25 cm**
- D. 50 cm**

CBSE 2015, 2019, 2020, 2021 (Term 2), 2024 (Sample Paper)

Q. Define the term power of accommodation of human eye. Write the name of the part of eye which plays a major role in the process of accommodation and explain what happens when human eye focuses
(i) nearby objects and
(ii) distant objects.

Ans. The ability of the eye lens to adjust its focal length, so as to clearly focus rays coming from distant as well as near objects on the retina, is called the power of accommodation of the eye. Ciliary muscles play a major role in maintaining power of accommodation.

(i) While focusing on nearby objects ciliary muscles contract, eye lens becomes thick and its focal length decreases. (ii) While focusing on distant objects ciliary muscles relax, eye lens becomes thin and its focal length increases.

DEFECTS OF VISION AND THEIR CORRECTION

1. Myopia (Near-sightedness)

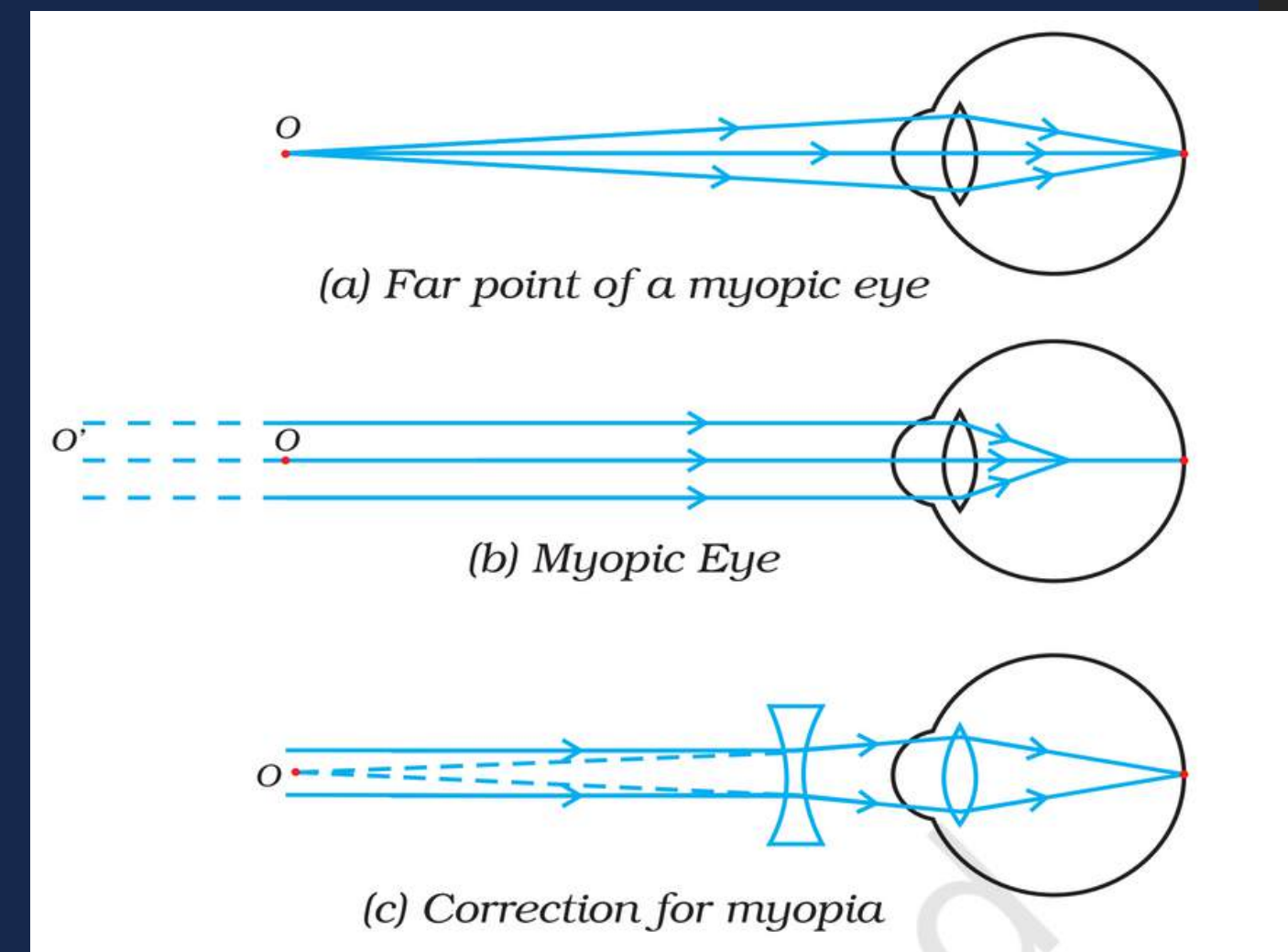
A person with myopia can see nearby objects clearly, but has difficulty in seeing distant objects. In this defect, the image of a distant object is formed in front of the retina instead of on it. The far point of the eye shifts closer than infinity.

Causes:

1. Excessive curvature of the eye lens
2. Elongation of the eyeball

Correction:

Myopia is corrected using a concave lens (diverging lens). It spreads out the incoming light rays so that they focus directly on the retina, allowing the person to see distant objects clearly.



2. Hypermetropia (Far-sightedness)

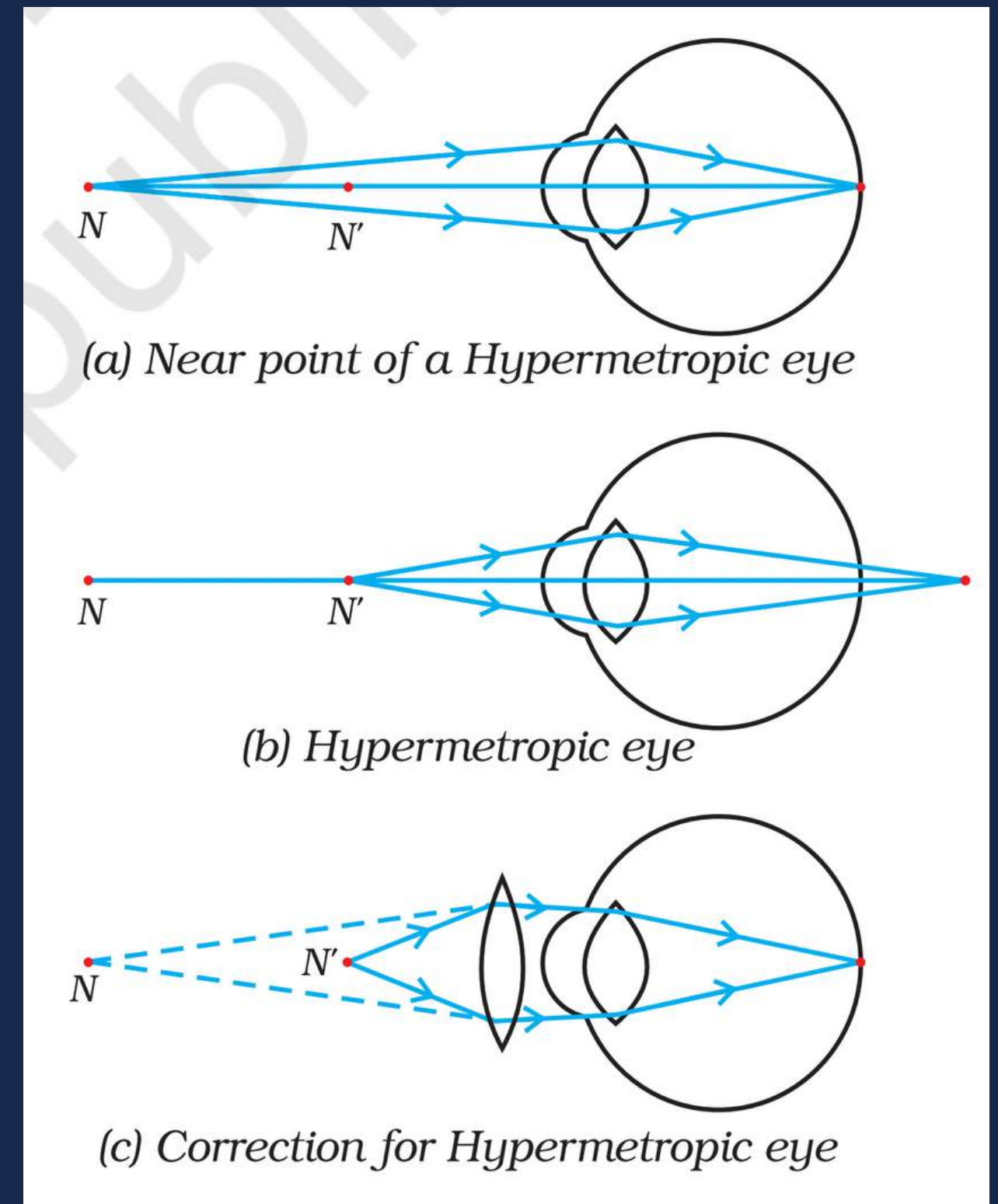
In hypermetropia, a person can see distant objects clearly, but cannot see nearby objects properly. This happens when the image of a nearby object is formed behind the retina. The near point of the eye is farther than the normal 25 cm.

Causes:

1. Focal length of the eye lens is too long
2. Eyeball is too short

Correction:

Hypermetropia is corrected using a convex lens (converging lens). It bends the incoming rays inward so they focus on the retina, allowing clear vision for nearby objects.



CBSE 2019, 2020 (Sample Paper), 2021 (Term 2), 2023 (Pre-board)

Q. Ravi, a 14-year-old student, started having difficulty seeing the whiteboard clearly from the back benches. He complained of headaches and eye strain while watching distant objects like street signs. His parents took him to an eye specialist, who diagnosed him with myopia and prescribed suitable lenses.

Questions:

- (a) What is myopia and what causes it?
- (b) Why can Ravi see nearby objects clearly but not distant ones?
- (c) What kind of lens is prescribed to Ravi and why?
- (d) With the help of a labelled diagram, explain how myopia can be corrected.
- (e) Suggest any two lifestyle tips that could help prevent or reduce the progression of myopia.

CBSE 2016, 2018, 2020 (Sample Paper), 2021 (Term 2), 2023 (Pre-board), 2024 (Sample Paper)

Q. Name the type of defect of vision a person is suffering from, if he uses convex lenses in his spectacles for the correction of his vision. If the power of the lenses is $+0.5\text{ D}$, find the focal length of the lenses.

CBSE 2016, 2018, 2020, 2021 (Term 2), 2023 (Pre-board)

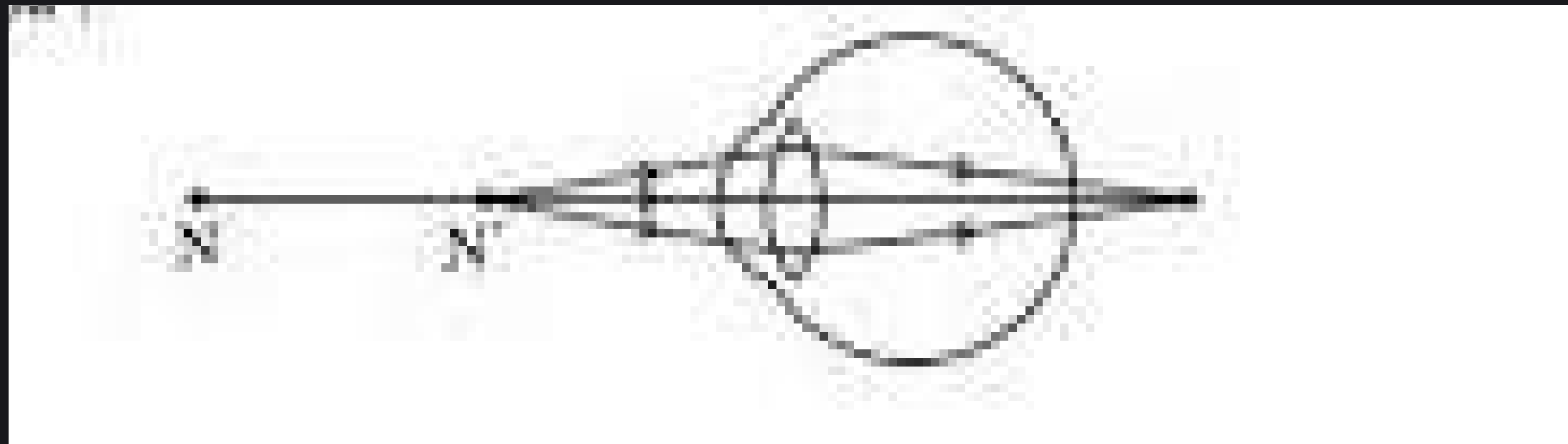
Q. Priya is unable to read books properly and feels the need to keep them far to see clearly. However, she can see distant objects easily. Her doctor recommends convex lenses.

- (a) What defect does Priya have?
- (b) Mention two causes of this defect.
- (c) Which type of lens is used for correction?
- (d) Draw ray diagram for its correction.

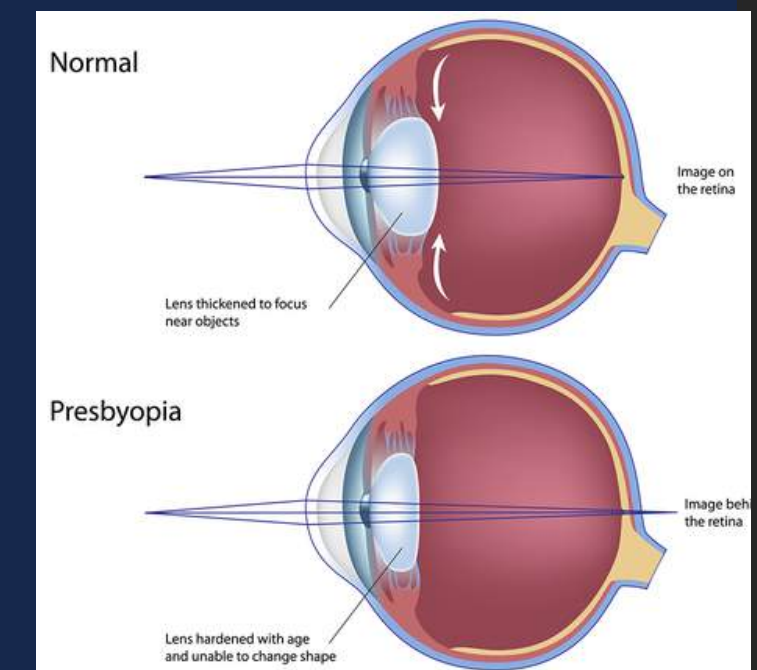
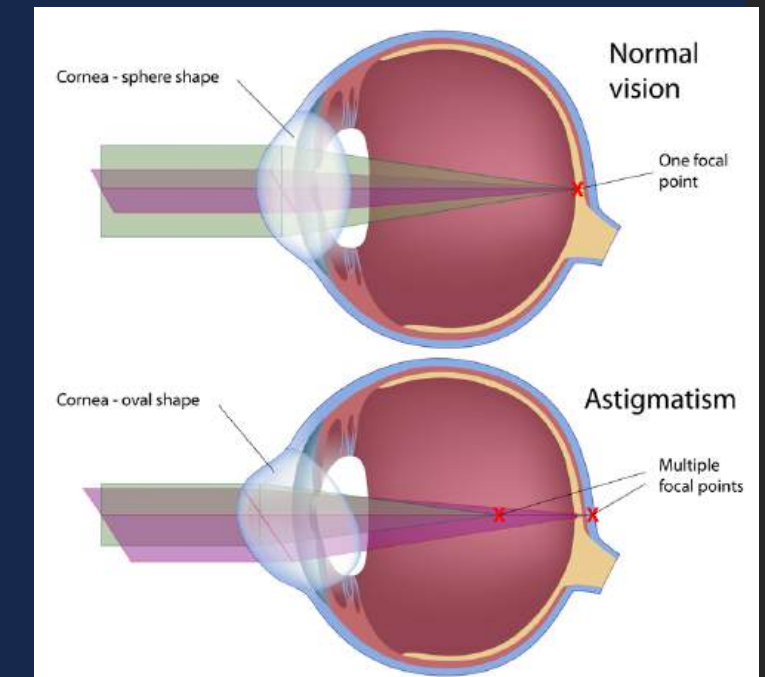
CBSE 2016, 2018, 2019, 2020 (Sample Paper), 2021 (Term 2), 2023 (Pre-board), 2024 (Sample Paper)

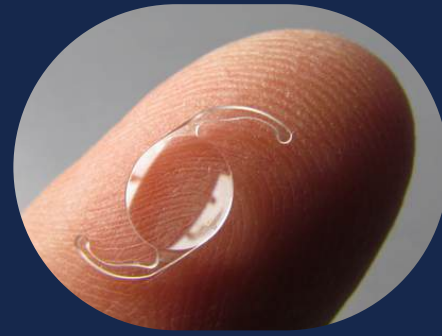
Q. Study the diagram given below and answer the questions that follow:

- (i) Name the defect of vision represented in the diagram. Give reason for your answer.
- (ii) List two causes of this defect.
- (iii) With the help of a diagram show how this defect of vision is corrected.



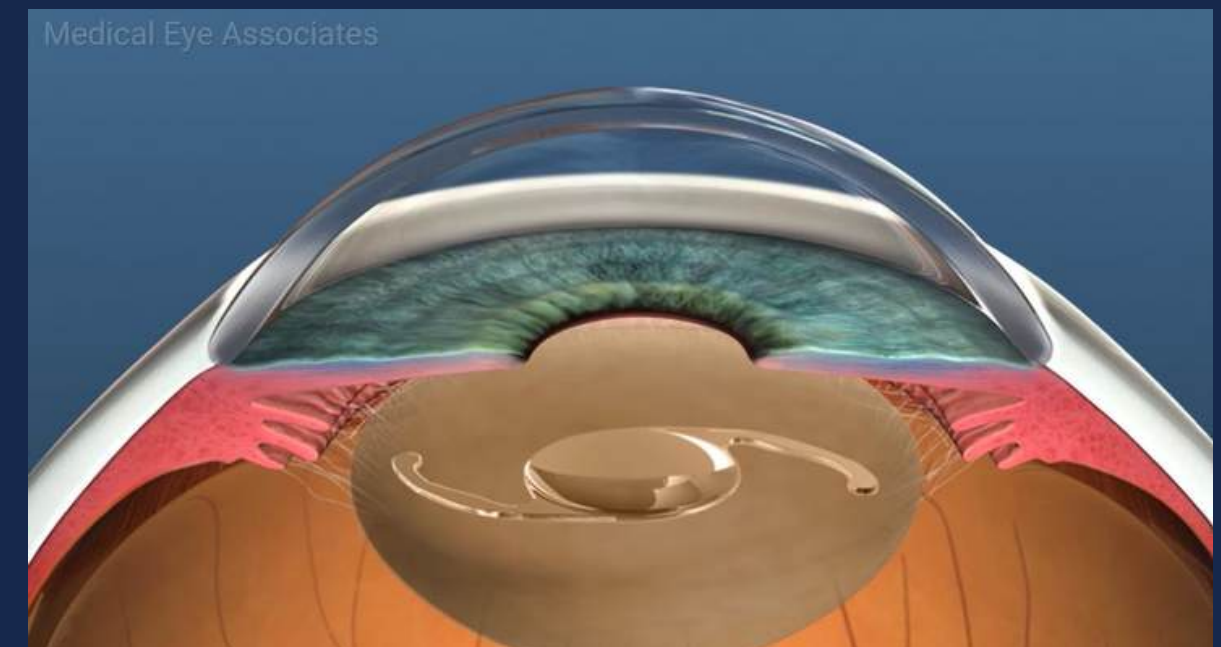
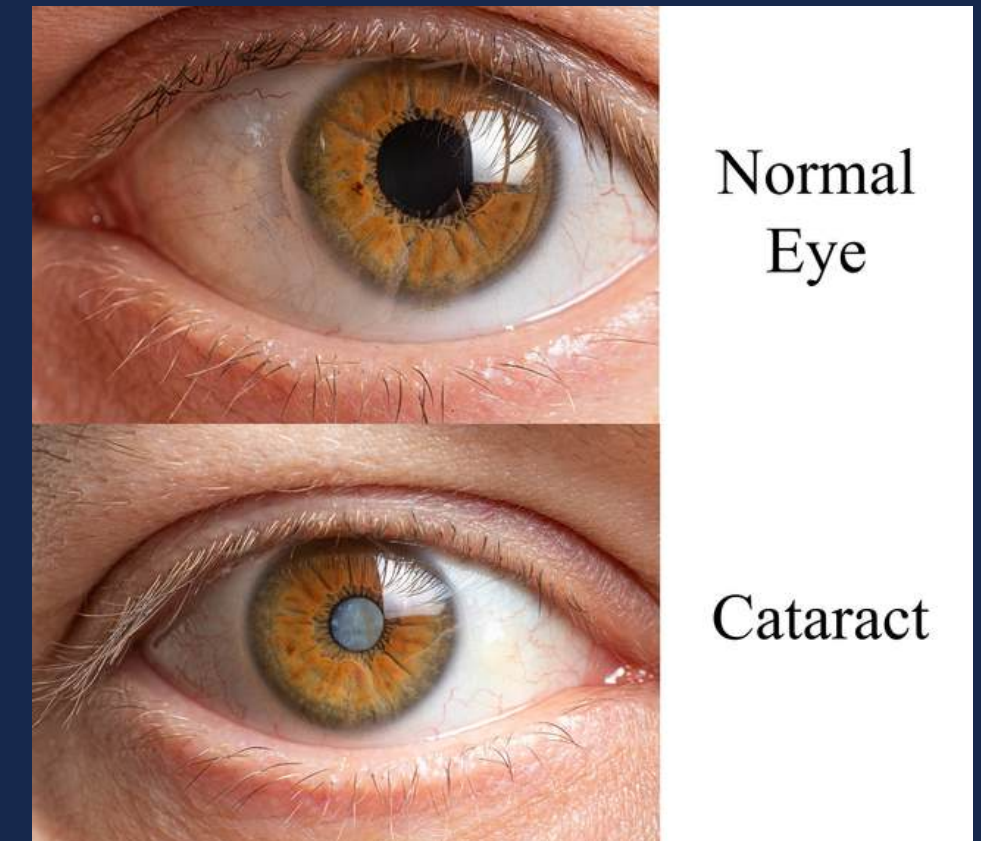
Condition	Description	Cause	Correction
Presbyopia	Old-age vision loss: Difficulty in seeing nearby objects clearly (near point recedes with age): often accompanied by hypermetropia.	(i) Weak ciliary muscles (ii) Decreased flexibility of eye lens	Use bifocal lenses {Upper part → Concave (for distant vision), Lower part → Convex (for near vision)}, contact lenses, or surgery
Astigmatism	Blurred or distorted vision at all distances due to an irregularly curved cornea or lens.	Uneven curvature of the cornea or lens surface.	Use of cylindrical lenses.





CATARACT

- *A condition in which crystalline lens of eye becomes milky and cloudy due to growth of membrane over it.*
- *It generally occurs in older adults.*
- *This causes partial or complete loss of vision.*
- *Vision can be restored through cataract surgery.*



CBSE 2020 (Sample Paper), 2021 (Term 1), 2023

Q. Why does vision become cloudy in old age?

Ans. Vision becomes cloudy in old age due to the gradual loss of flexibility of the eye lens and weakening of ciliary muscles. This condition is known as Presbyopia, where the eye is unable to focus on nearby objects clearly. The lens cannot adjust its shape easily, leading to blurred or cloudy near vision.

CBSE 2020 (Sample Paper), 2021 (Term 1), 2022 (Sample Paper), 2023 (Pre-board), 2024 (Sample Paper)

Mr. and Mrs. Sharma, both in their 50s, have recently developed difficulty in viewing both near and far objects. Their doctor informed them that this is a natural defect called presbyopia, commonly occurring with age. They were prescribed bifocal lenses.

Questions:

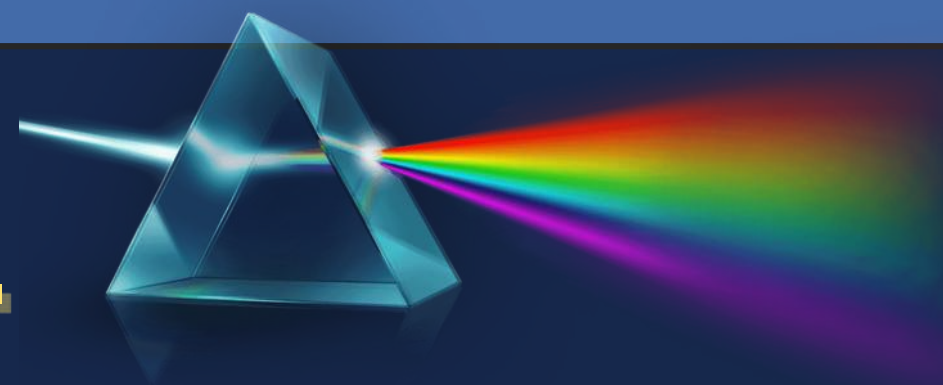
- (a) What is presbyopia? How is it different from hypermetropia?
- (b) What changes occur in the eye that lead to presbyopia?
- (c) How do bifocal lenses help in correcting presbyopia?
- (d) Can a person have both myopia and hypermetropia together? What is the solution in such cases?

CBSE 2021 (School Pre-board), 2023 (Sample Paper)

Q. Which defect of vision is caused due to uneven curvature of the cornea?

- A. Myopia
- B. Hypermetropia
- C. Presbyopia
- D. Astigmatism

PRISM

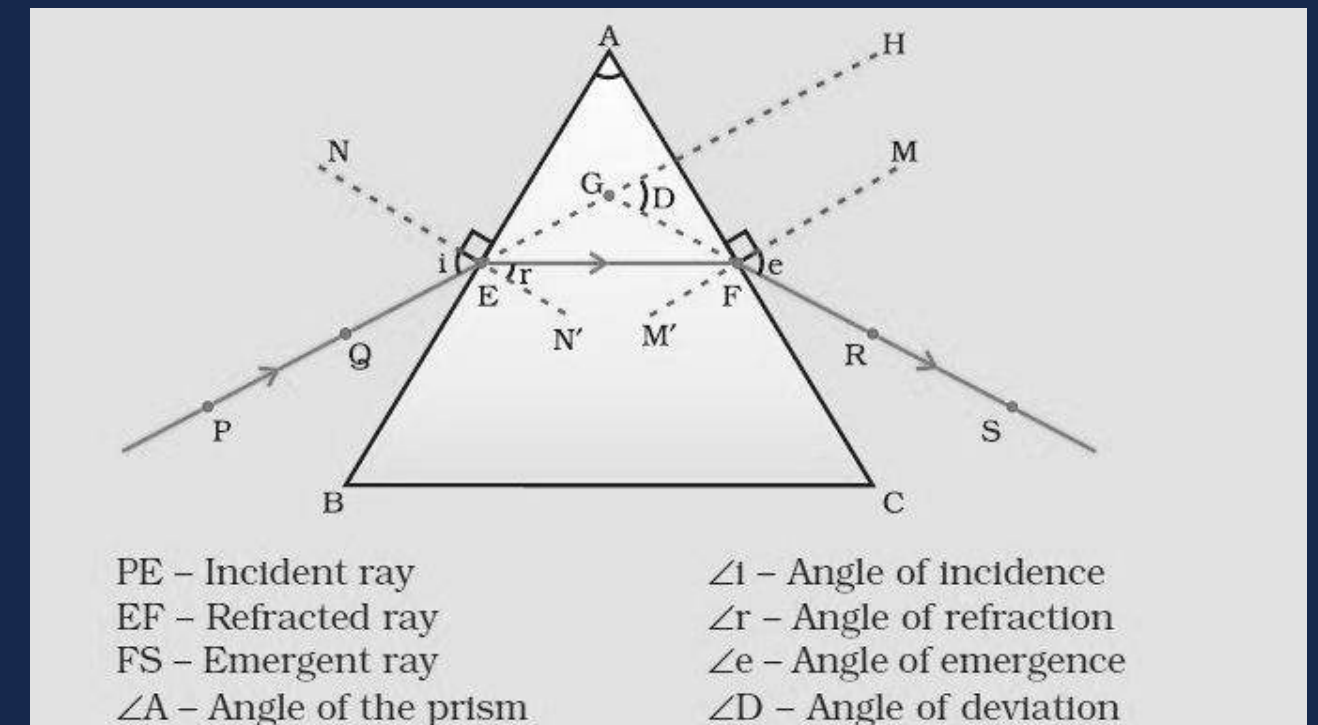


Prism is a transparent refracting medium bounded by at least two lateral surfaces, inclined at each other at a certain angle.

It consists of two triangular bases and three rectangular lateral surfaces.

REFRACTION OF MONOCHROMATIC LIGHT THROUGH A PRISM

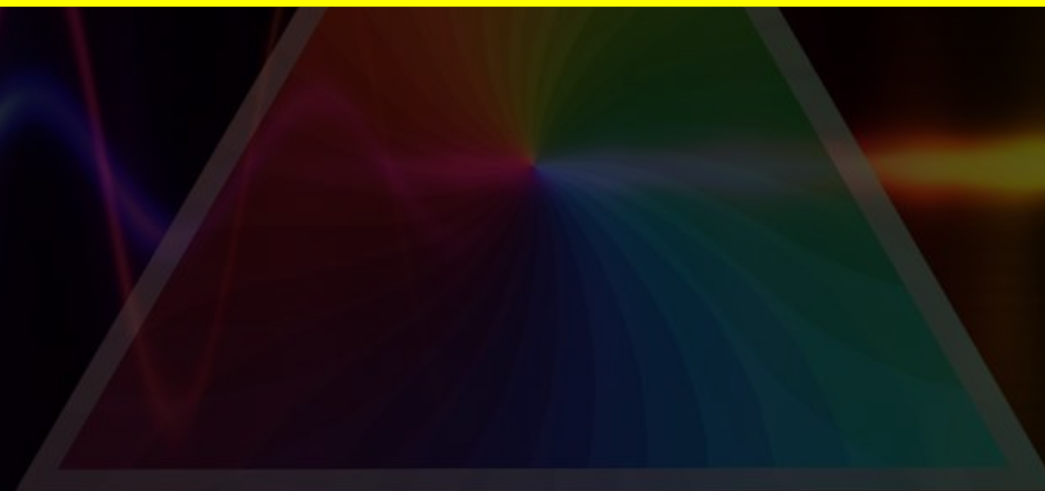
- **The angle between the two lateral faces of the prism is known as the angle of the prism (A).**
- **The angle of deviation is the angle between the direction of the incident ray and the direction of the emergent ray after light passes through a prism.**
- **Refraction in Prism occurs at two surfaces:**
 - 1. From air to glass → bends towards normal**
 - 2. From glass to air → bends away from normal**



CBSE 2022 (Board), 2023 (Pre-board), 2024 (Sample Paper)

Q. When a ray of light passes through a prism, it bends:

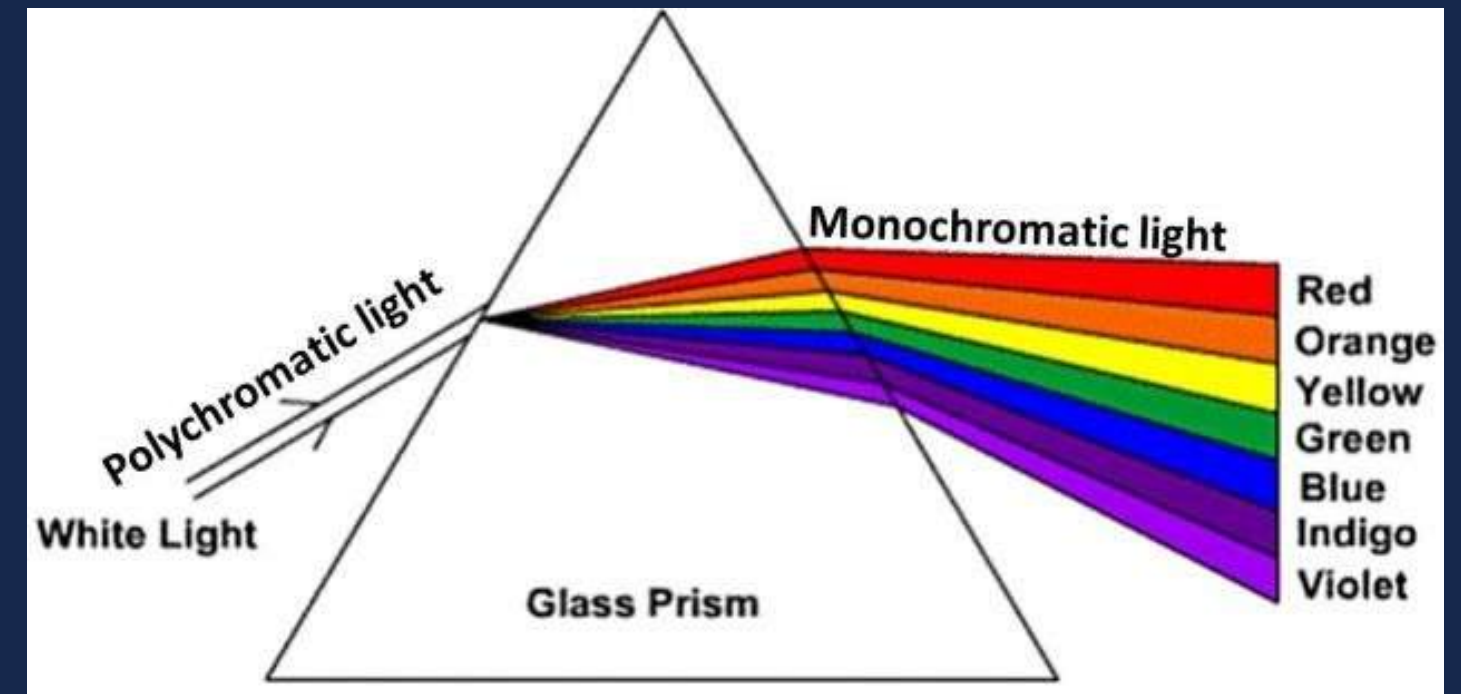
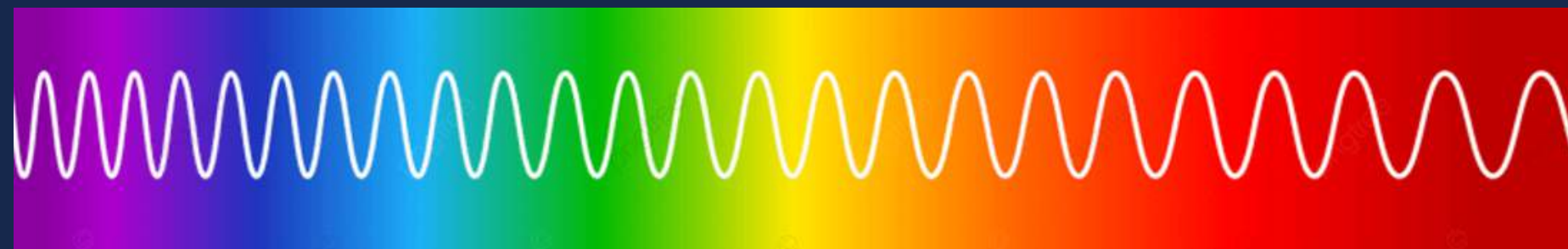
- A) Towards the base
- B) Towards the apex
- C) Along the normal
- D) Undeviated



DISPERSION OF WHITE LIGHT BY PRISM

The phenomenon of splitting of white light into its constituent colours when it passes through a prism is called dispersion.

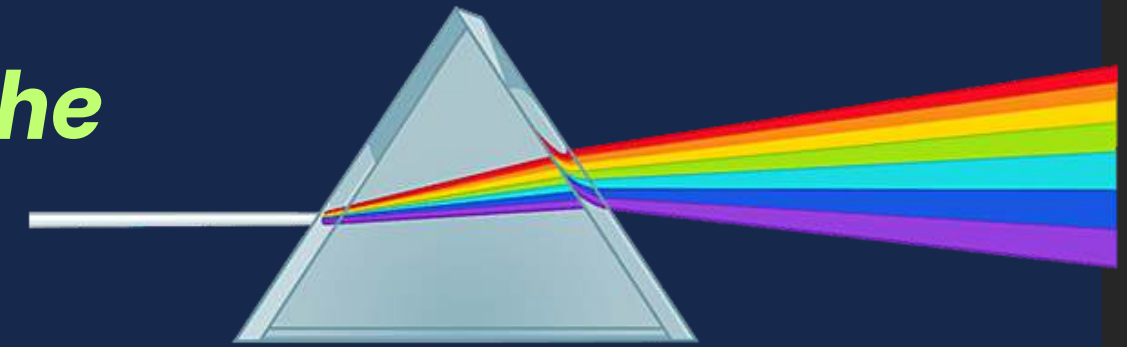
*This band of seven colors so obtained the **VIBGYOR** (violet, indigo, blue, green, yellow, orange, red) is called a spectrum.*



The process of dispersion happens only due to the prism's angled surfaces, which are not parallel like in a rectangular glass slab. This unique bending at different angles causes white light to spread out into a colourful band called the spectrum.

CAUSE OF DISPERSION

In a vacuum and air, light rays of all colors travel at the same speed.



However, in other mediums, they travel at different speeds and bend at different angles, causing light to disperse into its constituent colors.

Red light has the maximum wavelength, and violet light has the minimum wavelength.

*Thus, in any medium, **red light** travels the fastest and deviates the least, while **violet light** travels the slowest and deviates the most, i.e.,*

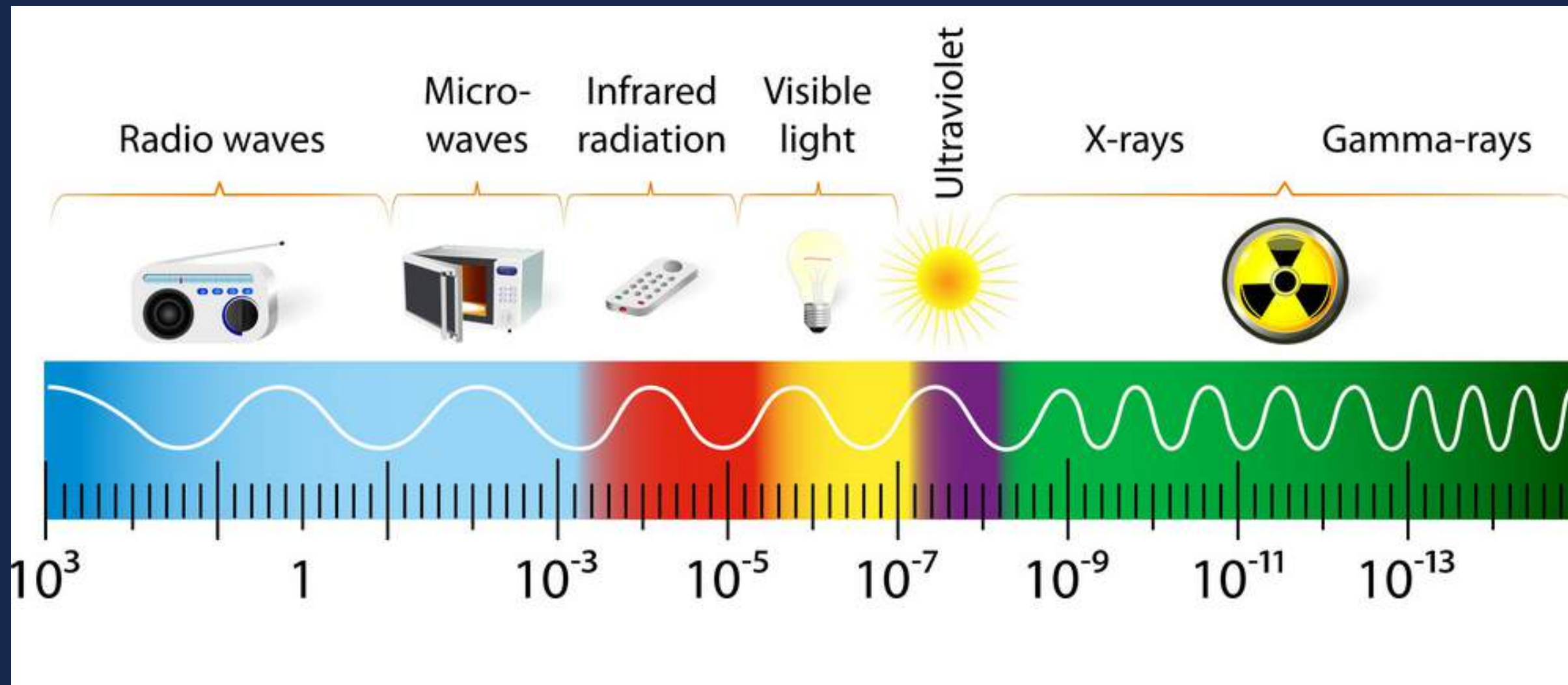
$$\text{Wavelength} \propto \text{Velocity} \propto 1/\text{Deviation}$$

ELECTROMAGNETIC SPECTRUM

The visible light range is from 400 nm to 700 nm in wavelength.

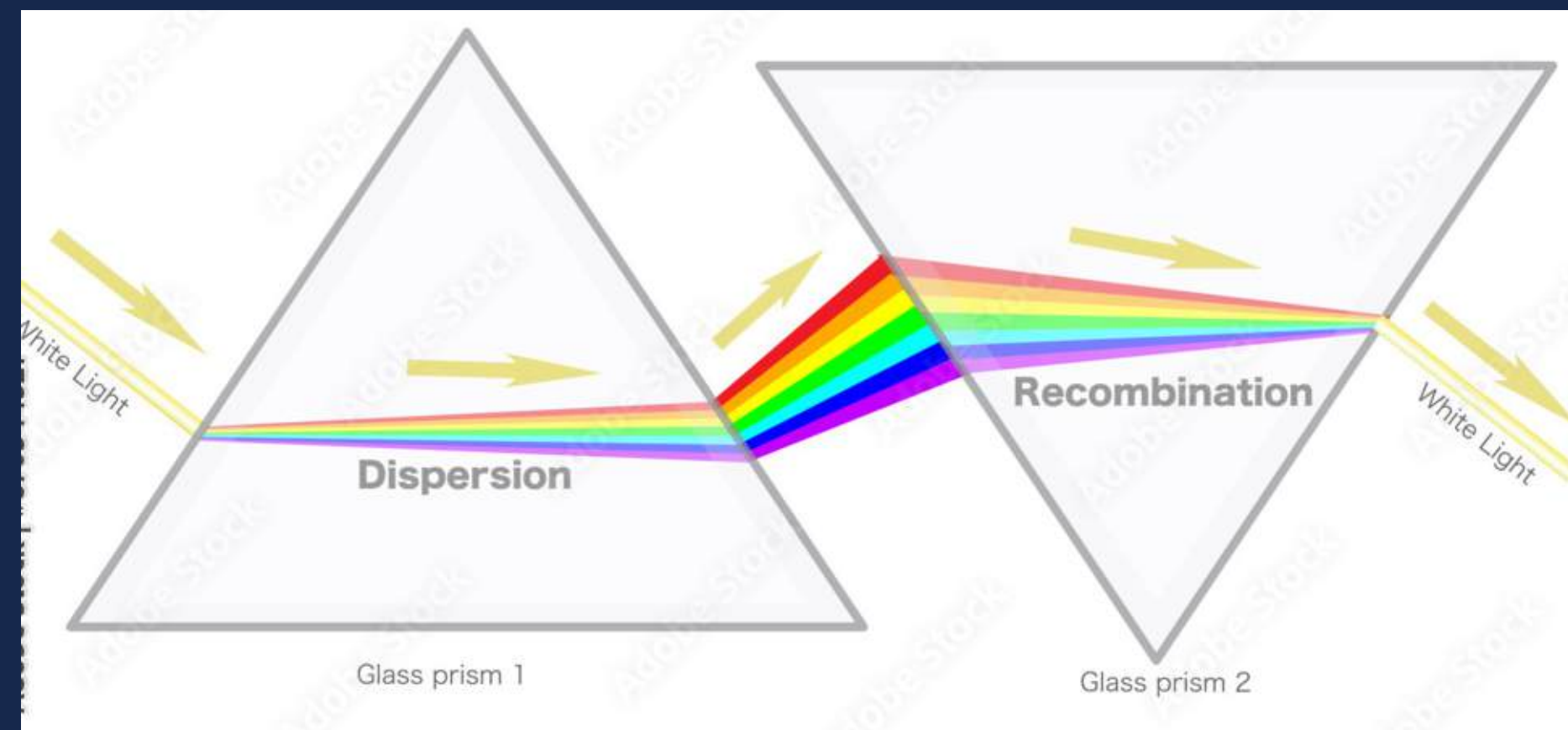
Violet: 400 nm (shortest wavelength)

Red: 700 nm (longest wavelength)



NEWTON'S PRISM EXPERIMENT

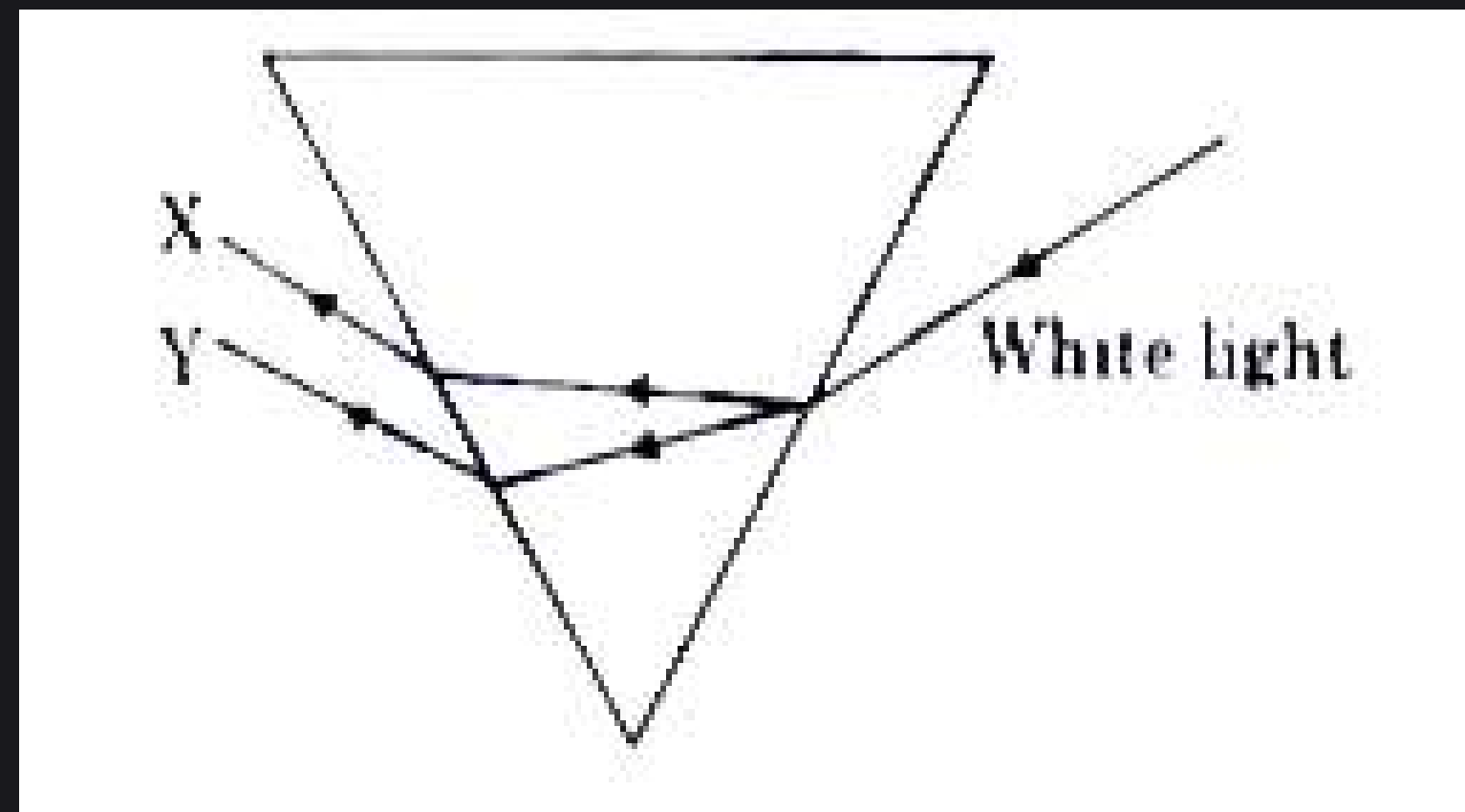
Sir Isaac Newton showed that sunlight is made of seven colours by passing it through a prism to form a spectrum. He then used a second inverted prism to recombine the colours back into white light, proving that white light is a mixture of all seven colours.



CBSE 2012, 2016 (Pre-board), 2018, 2020 (Sample Paper), 2021 (Term 1), 2022 (Term 2), 2023 (Pre-board), 2024 (Sample Paper)

Q. In the diagram given below, X and Y are the end colours of the spectrum of white light. The colour of 'Y' represents the:

- (A) Red
- (B) Violet
- (C) Blue
- (D) Yellow



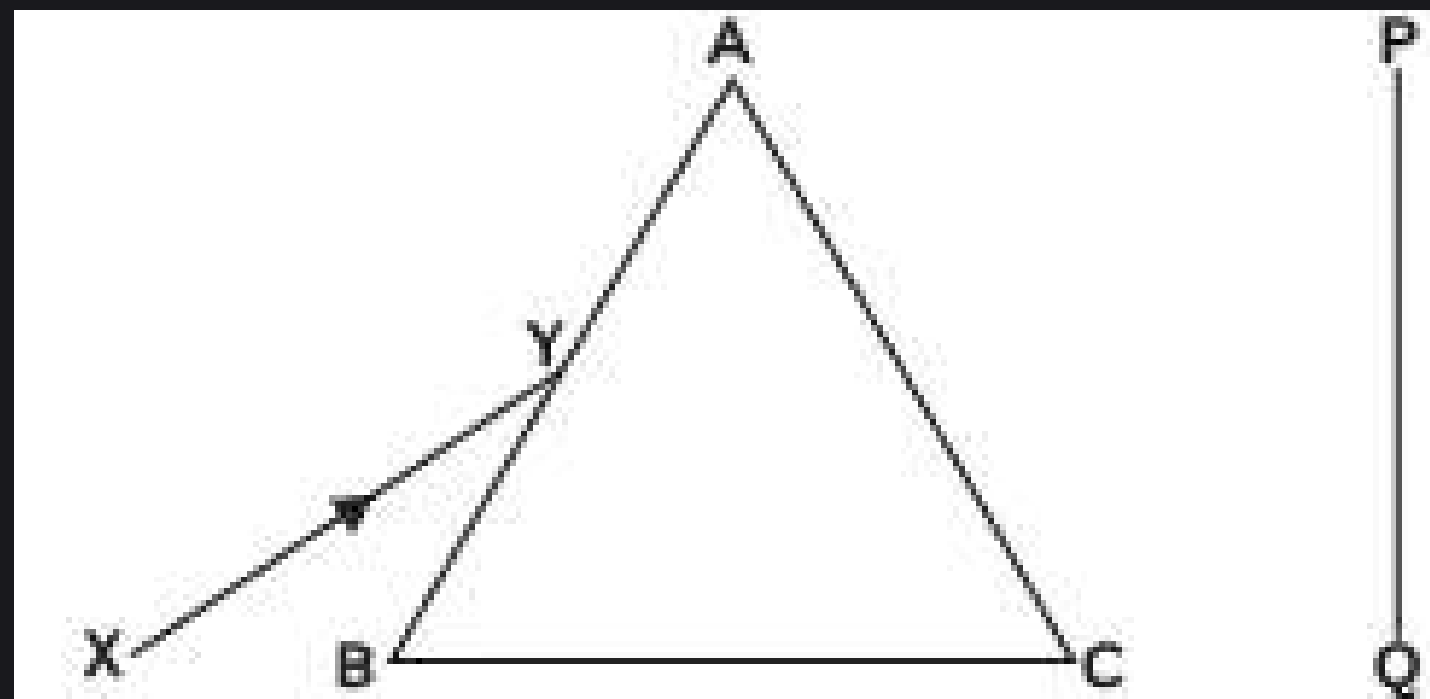
CBSE 2015, 2016 (Pre-board), 2018, 2020 (Sample Paper), 2021, 2022 (Sample Paper), 2023 (Pre-board), 2024 (Sample Paper)

Q. Which colour deviates the most and the least in dispersion through a prism, and why?

- (a) Violet deviates the most; red deviates the least due to difference in speed in glass
- (b) Red deviates the most; violet deviates the least due to same wavelength
- (c) Violet deviates the most; red deviates the least due to equal refractive index
- (d) Blue deviates the most; red deviates the least due to scattering

CBSE 2016, 2017, 2018, 2020 (Sample Paper), 2021 (Term 2), 2023 (Pre-board)

Q. A narrow beam XY of white light is passing through a glass prism ABC as shown in the diagram: Trace it on your answer sheet and show the path of the emergent beam as observed on the screen PQ. Name the phenomenon observed and state its cause.



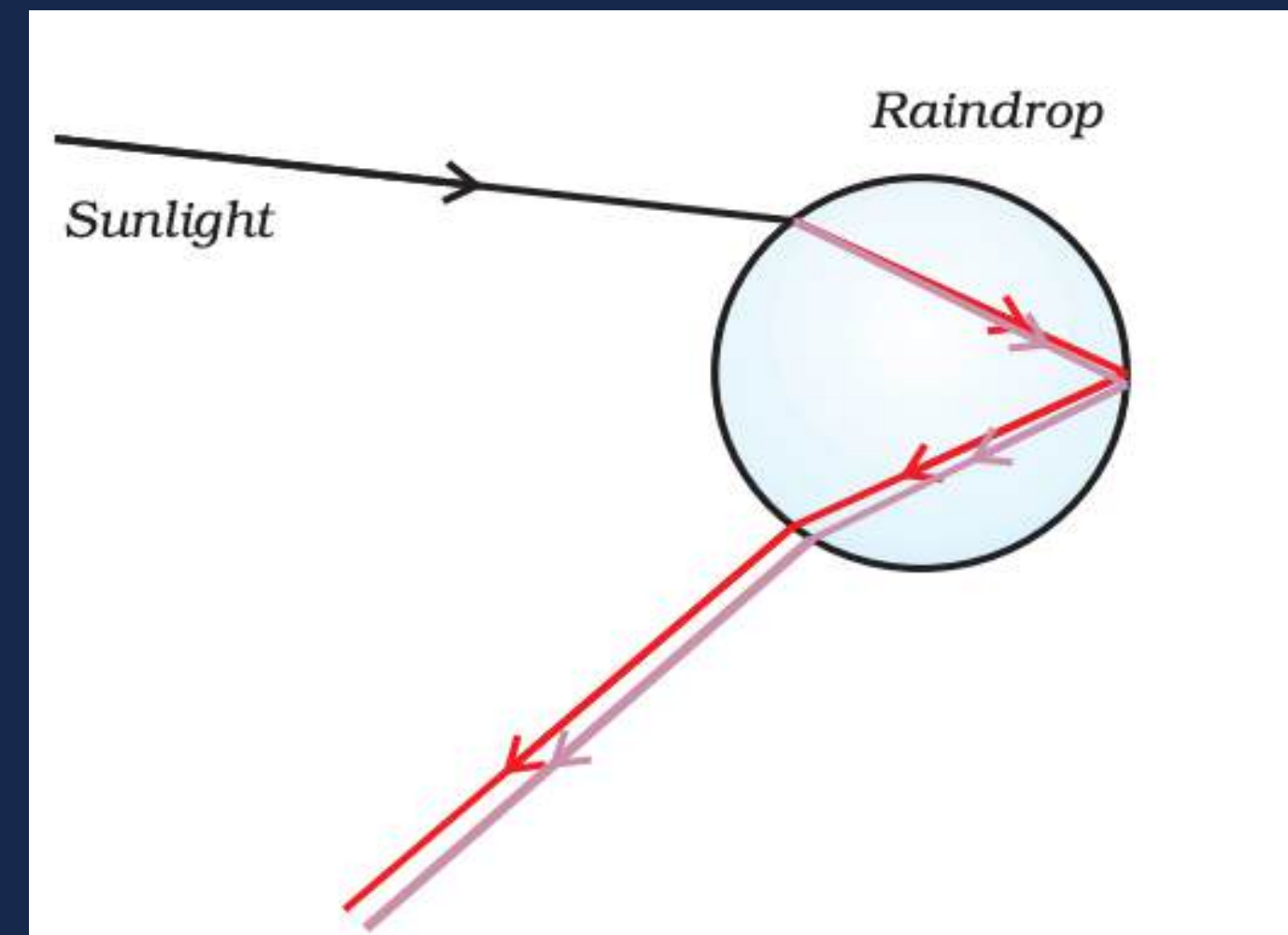
RAINBOW FORMATION

A rainbow is a natural example of dispersion. It forms in the sky after rainfall when the Sun is shining.

Tiny water droplets in the atmosphere act like small prisms. These droplets:

- 1. Refract sunlight when it enters,*
- 2. Disperse it into seven colours,*
- 3. Internally reflect the light inside the droplet,*
- 4. Then refract again when light exits the droplet.*

Due to this, the seven colours reach the observer's eye, and we see a beautiful rainbow. A rainbow is always formed in the opposite direction of the Sun.



CBSE 2015, 2017, 2019 (Pre-board), 2020 (Sample Paper), 2021 (Term 1), 2022 (Sample Paper), 2023 (Pre-board), 2024 (Sample Paper)

Q. The phenomena of light involved in the formation of a rainbow in the sky are

- (a) Refraction, dispersion and reflection
- (b) Refraction, dispersion and total internal reflection
- (c) Dispersion, scattering and reflection
- (d) Dispersion, refraction and internal reflection

CBSE 2015, 2017, 2019 (Pre-board), 2020 (Sample Paper), 2021 (Term 2), 2022 (Sample Paper), 2023 (Pre-board), 2024 (Sample Paper)

Q.Explain the formation of a rainbow and explain why a rainbow is always formed in a direction opposite to the Sun.

Ans. A rainbow is formed due to the dispersion, refraction, and internal reflection of sunlight by water droplets present in the atmosphere.

When sunlight enters a water droplet, it slows down and bends (→ refraction) and splits into 7 colours (→ dispersion).

These colours are: Violet, Indigo, Blue, Green, Yellow, Orange, Red (VIBGYOR).

The light is then internally reflected from the back of the droplet.

It again refracts when it comes out of the droplet, and the seven colours spread out to form a rainbow.

The combined effect of millions of such droplets forms a rainbow in the sky.

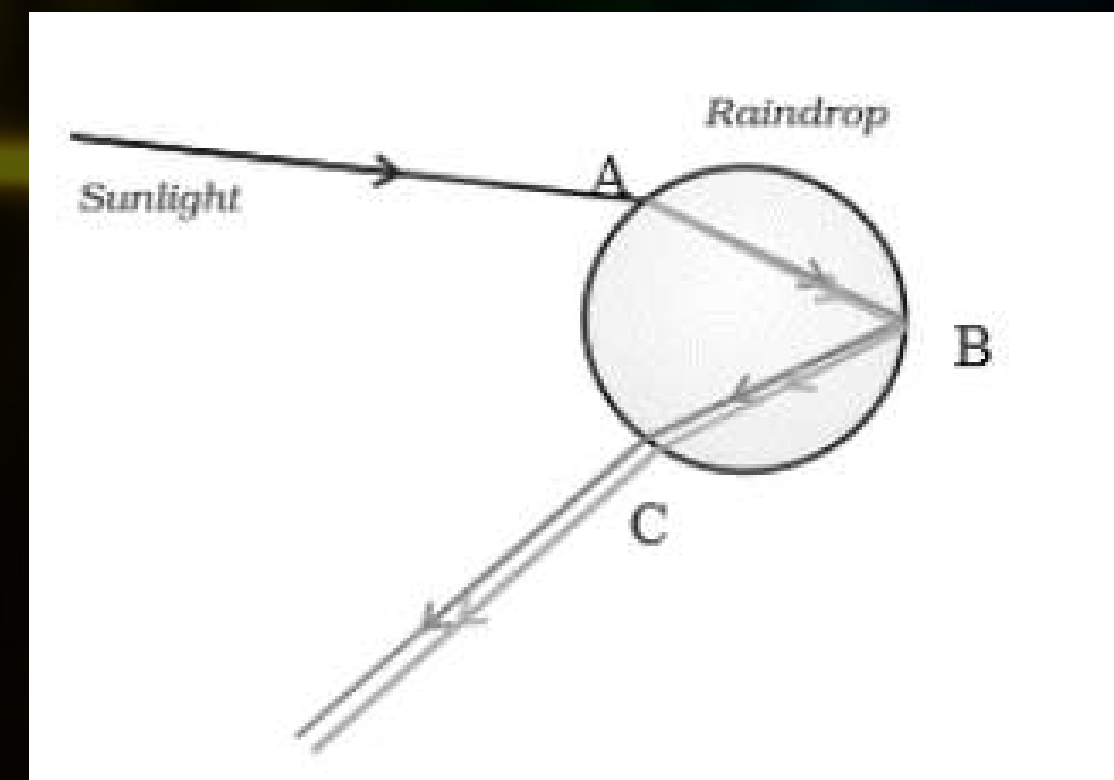
- A rainbow is always seen when the observer stands with their back to the Sun.*
- Sunlight enters the droplets in front of the observer, and the emerging dispersed light returns back toward the observer's eyes.*

CBSE 2020 (Sample Paper), 2021 (Term 2), 2023 (Pre-board), 2024 (Sample Paper)

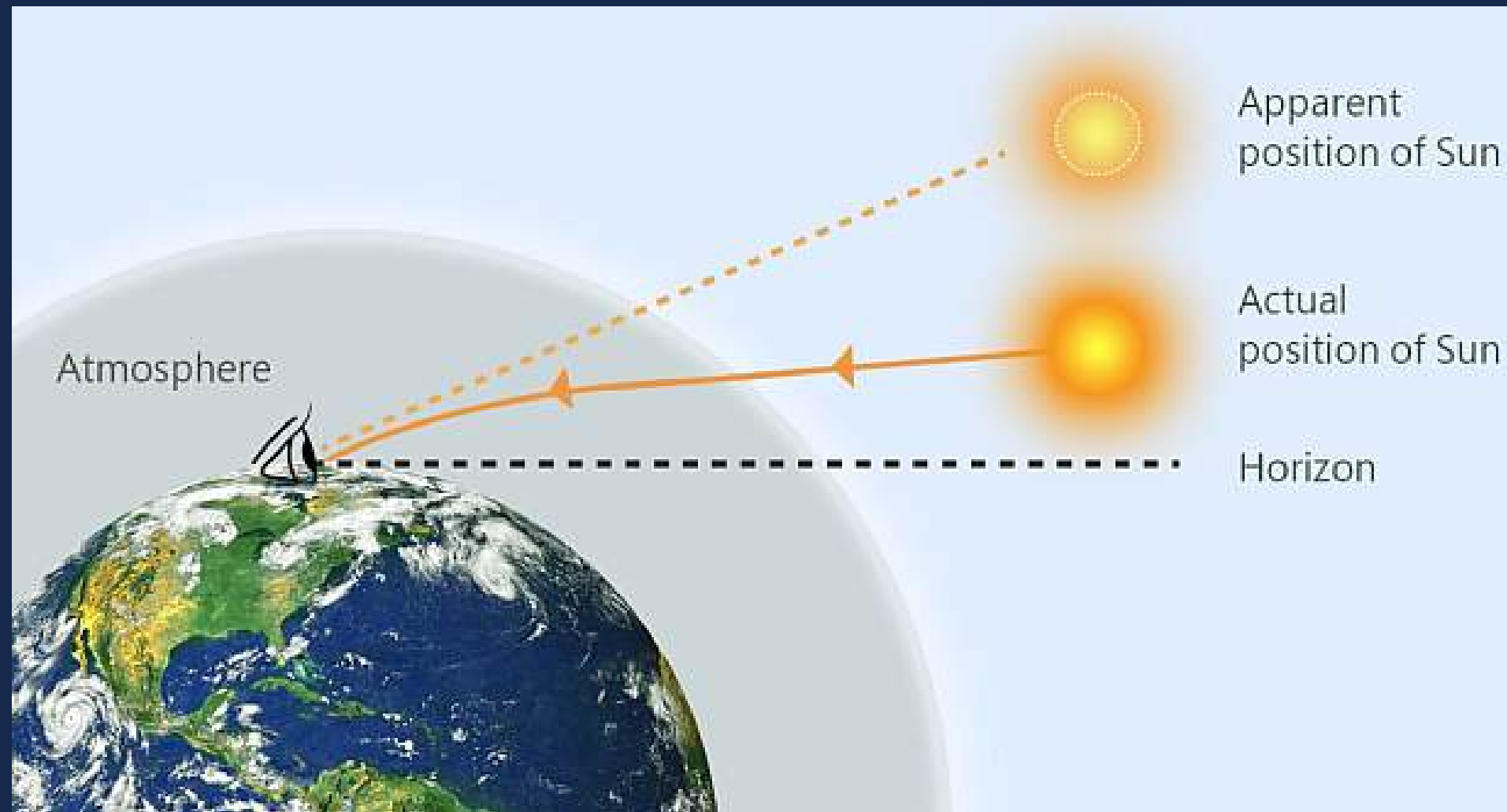
Q. Draw a ray diagram to show the formation of a rainbow in the sky. On this diagram mark A – where dispersion of light occurs, B – where internal reflection of light occurs and C – where refraction of light occurs. List two necessary conditions to observe a rainbow.

Ans. Two conditions:

- (i) The Presence of tiny water droplets in the atmosphere.
- (ii) Position of Sun at the back(behind) of the observer.



ATMOSPHERIC REFRACTION

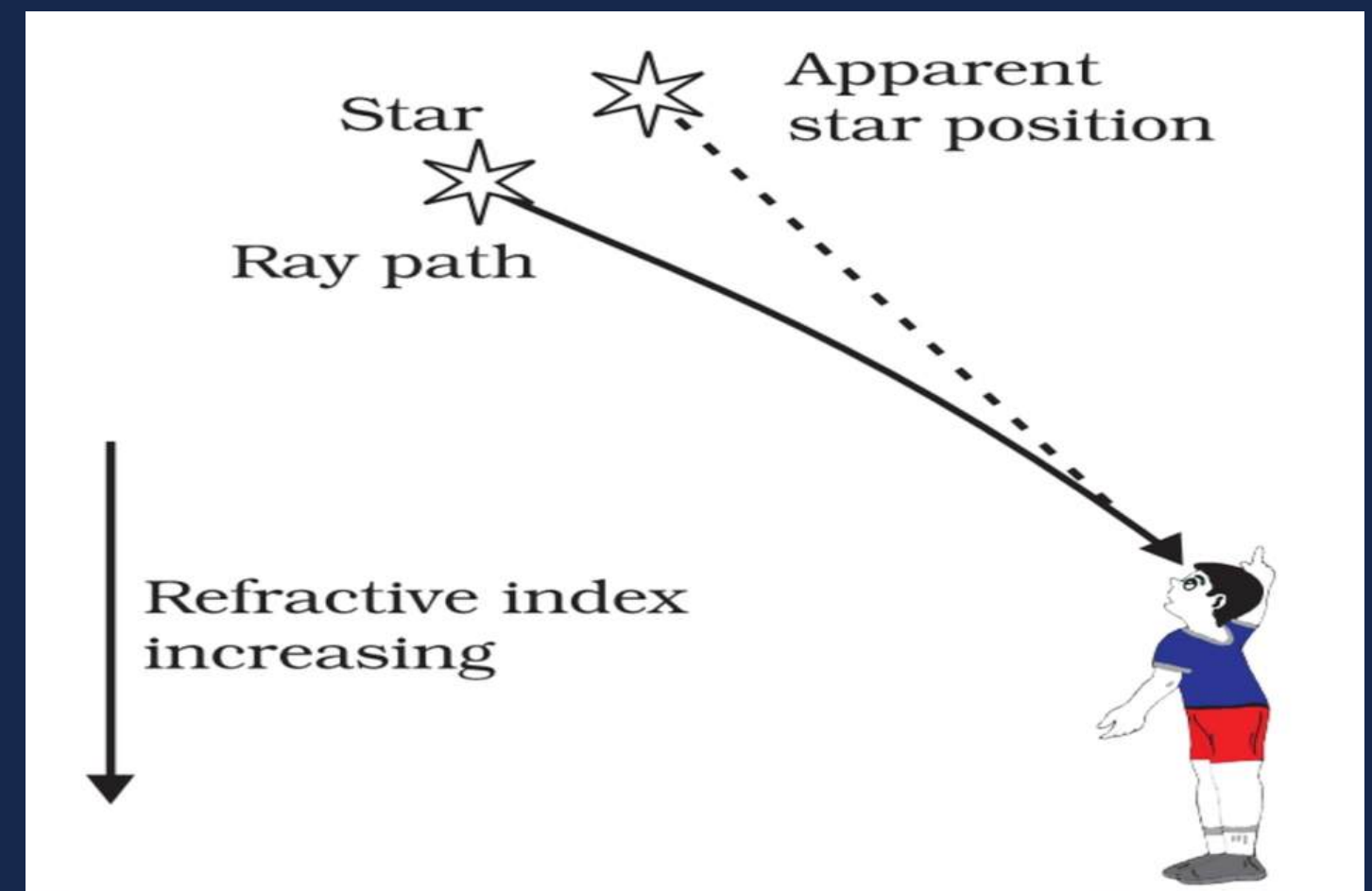


- *It is the bending of light or other electromagnetic waves as they pass through the Earth's atmosphere.*
- *It occurs because the atmosphere's layers have different optical densities, which is due to variations in air density with height.*

EFFECTS OF ATMOSPHERIC REFRACTION

Apparent Position of Stars

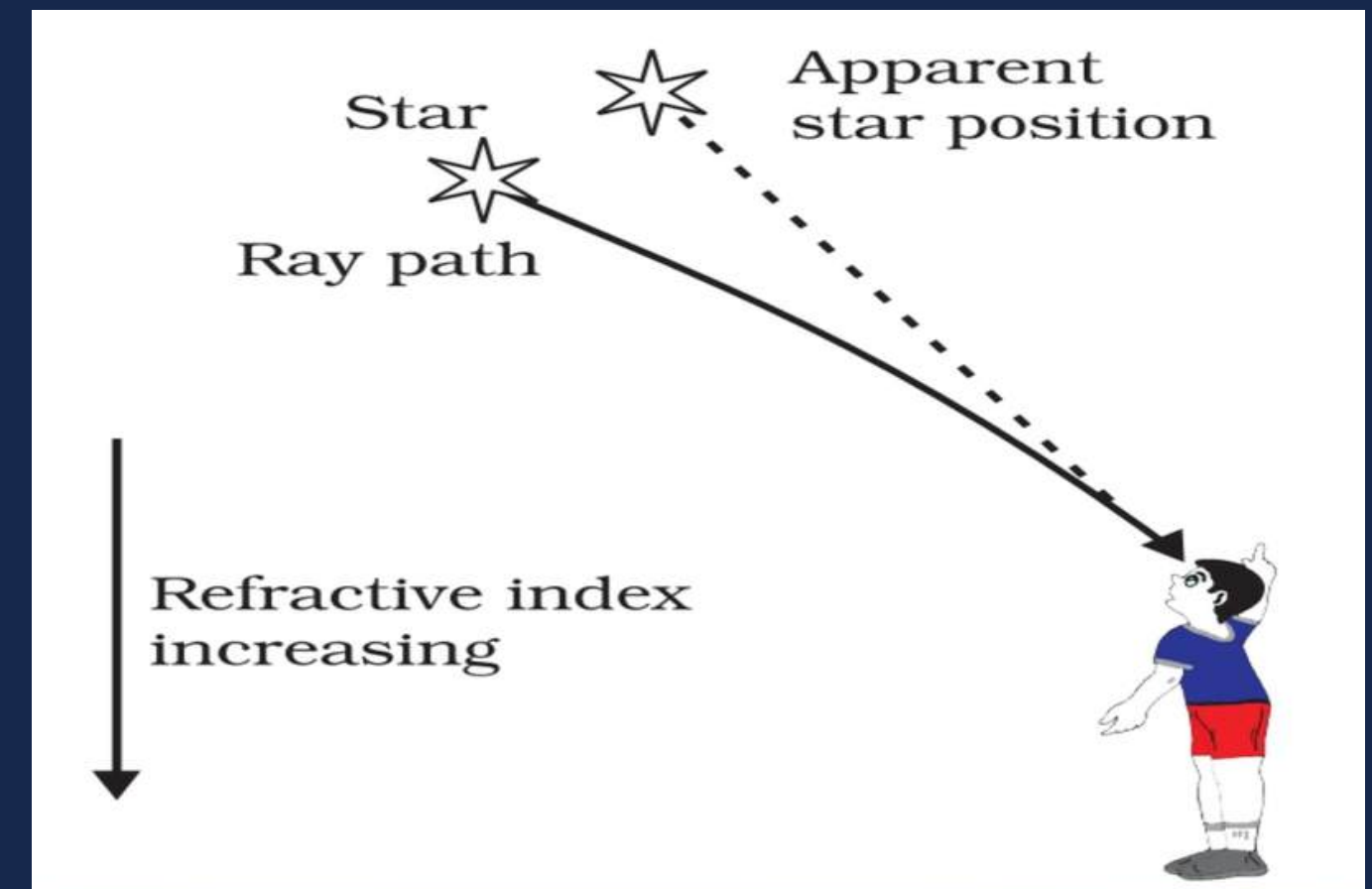
When starlight enters Earth's atmosphere, it passes through layers of air with different temperatures and densities, causing a gradual change in the refractive index. This leads to continuous refraction of light towards the normal, bending the light path. As a result, the star appears slightly higher than its actual position – this is called the apparent position of the star. The effect is more prominent near the horizon, where atmospheric refraction is stronger.



EFFECTS OF ATMOSPHERIC REFRACTION

Twinkling of stars

- Twinkling of stars is caused by atmospheric refraction. Due to continuous changes in air temperature, pressure, and density, the refractive index of the atmosphere keeps changing, causing the path of starlight to bend irregularly. Since stars appear as point sources of light, even a slight change in the light path leads to variation in brightness. This makes the star appear to twinkle.



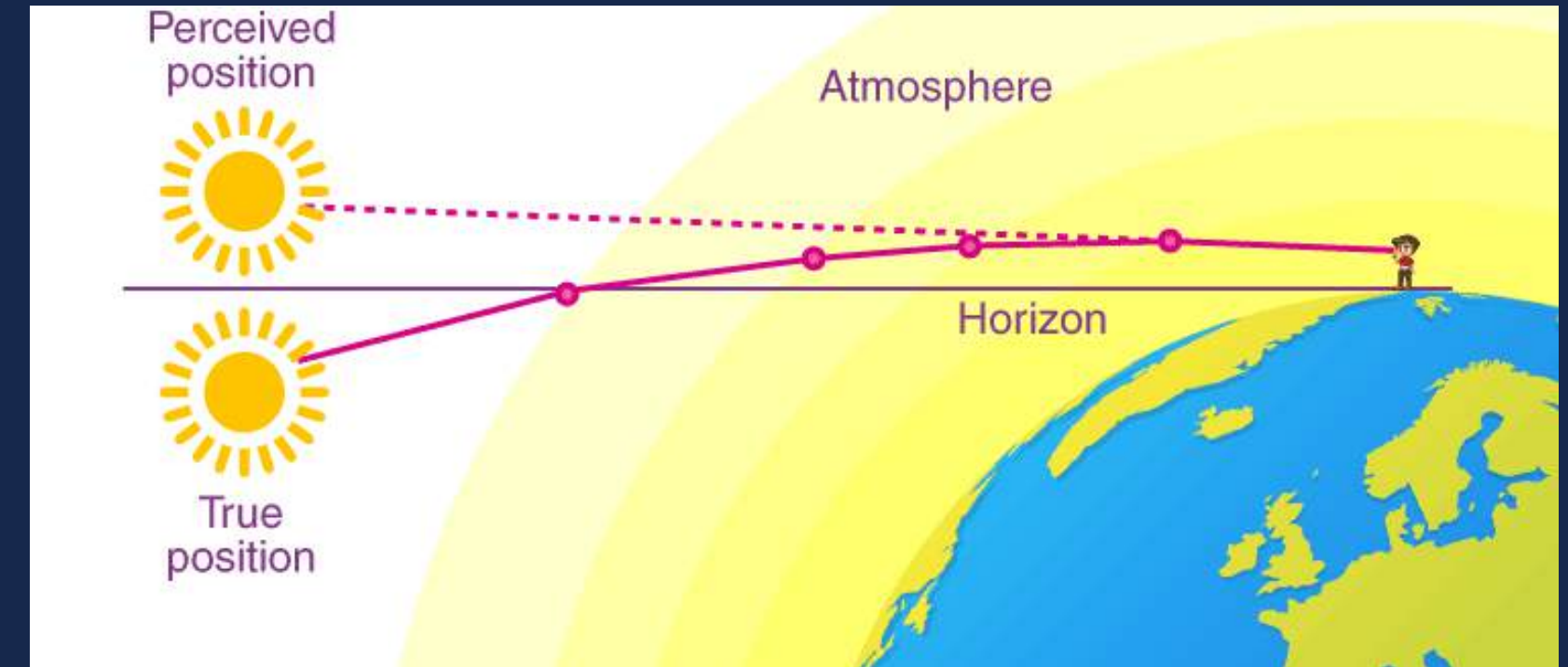
BUT WHY DO PLANETS NOT TWINKLE?

- *Planets are much closer to Earth than stars.*
- *They appear as extended sources of light, not point-sized.*
- *Each planet acts like a group of many tiny point sources.*
- *Atmospheric refraction affects each point slightly differently, but their light averages out.*
- *So the overall brightness remains stable. Hence, planets appear to shine steadily and do not twinkle.*



ADVANCED SUNRISE & DELAYED SUNSET

- *The Sun is visible to us about 2 minutes before the actual sunrise, and about 2 minutes after the actual sunset because of atmospheric refraction.*
- *By 'actual sunrise,' we mean the actual crossing of the horizon by the Sun.*



SCATTERING OF LIGHT

The reflection of light from an object in all directions is called scattering of light. It depends on type of particle.

- ***Very fine particles scatter mainly blue light.***
- ***Large sized particle scatter light of longer wavelength.***
- ***Shorter wavelength greater will be the scattering***



SCATTERING OF LIGHT

Tyndall effect

- *The Earth's atmosphere is a heterogeneous mixture containing smoke, tiny water droplets, dust, and air particles.*
- *When a beam of light strikes such fine particles, the path of the beam becomes visible.*
- *The light reaches us, after being reflected diffusely by these particles.*
- *It can also be observed when sunlight passes through a canopy of a dense forest.*

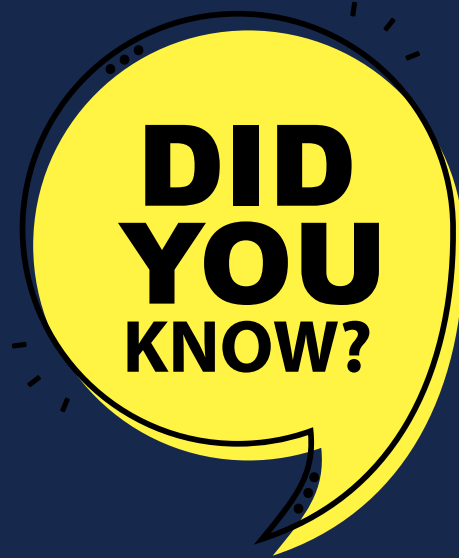


SCATTERING OF LIGHT

Blue Sky

- *Sunlight gets scattered by small air molecules and other fine particles in the atmosphere during its passage.*
- *Scattering is inversely proportional to wavelength, i.e. Blue with the shortest wavelength will scatter more compared to the red.*
- *This greater scattering of blue light by air molecules makes the sky appear blue on a clear day.*





- **Without Earth's atmosphere, no scattering of light would occur, and the sky would appear dark, similar to what passengers observe at high altitudes.**
- **Why red is used for danger signals: Red light scatters the least due to its longest wavelength, allowing it to remain visible over long distances through fog or smoke.**



CBSE 2021 (Term 1 MCQ), 2022 (Sample Paper), 2023 (Pre-board), 2024 (Sample Paper)

Q. On a windy night, stars appear to twinkle more. Why?

- A) More light scattering
- B) Light gets absorbed
- C) Greater variation in atmospheric refraction
- D) Stars move faster

CBSE 2020 (Sample Paper), 2021 (Term 1), 2022 (Sample Paper), 2023 (Pre-board), 2024 (Sample Paper)

Q. Assertion (A): Stars appear to twinkle in the sky.

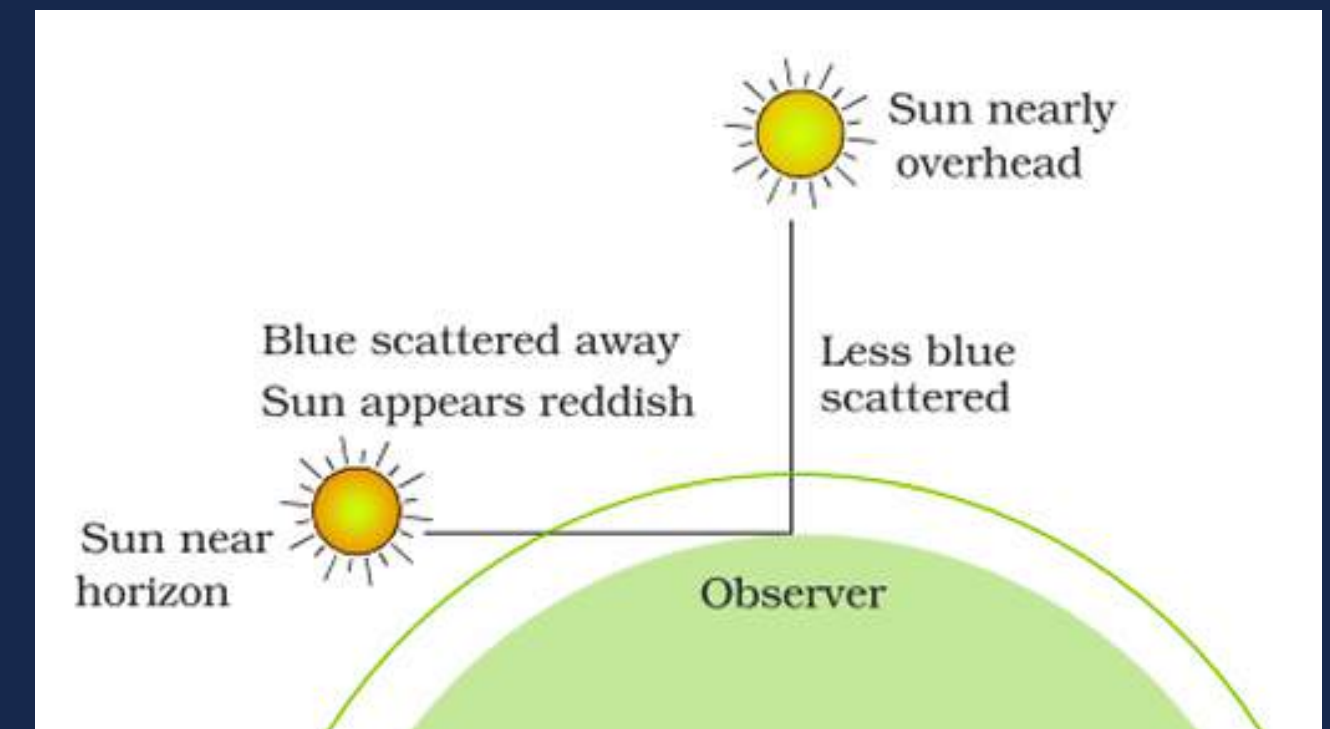
Reason (R): Atmospheric refraction causes the light of stars to bend continuously.

- (a) Both A and R are true, and R is the correct explanation of A
- (b) Both A and R are true, but R is not the correct explanation of A
- (c) A is true, but R is false
- (d) A is false, but R is true

SCATTERING OF LIGHT

Color of Sun at Sunrise and sunset

- The Sun appears red during sunrise and sunset because its light travels a longer distance through the Earth's atmosphere.
- *Shorter wavelengths (blue and violet) are scattered away, while longer wavelengths (red and orange) dominate, giving the Sun a red appearance.*



CBSE 2015, 2017, 2020 (Sample Paper), 2021 (Term 1), 2022 (Sample Paper), 2023 (Pre-board), 2024 (Sample Paper)

Q. When a beam of white light passes through a region having very fine dust particles, the colour of light mainly scattered in that region is:

- (a) red
- (b) orange
- (c) blue
- (d) yellow

CBSE 2015, 2016, 2018, 2019, 2020 (Sample Paper), 2021 (Term 2), 2023 (Pre-board), 2024 (Sample Paper)

Q. Define atmospheric refraction. Explain the following effects with reason:

- (i) Twinkling of stars
- (ii) Advanced sunrise and delayed sunset

CBSE 2016, 2018, 2019, 2020 (Sample Paper), 2021 (Term 2), 2023 (Pre-board), 2024 (Sample Paper)

Q. While watching a sunset, Arjun observed that the Sun appeared oval in shape and stayed visible for some time even after it had gone below the horizon.

1: What phenomenon is responsible for this?

2: Why does the Sun look oval?

3: How long does atmospheric refraction affect the visibility of the Sun?

Ans. 1. Atmospheric refraction is responsible for this observation.

2. The Sun looks oval because the lower part is more refracted than the upper part due to varying air densities, causing unequal bending of light.

3. Atmospheric refraction affects the visibility of the Sun for about 2 minutes after sunset.

CBSE 2015, 2017, 2020 (Sample Paper), 2021 (Term 1), 2023 (Board)

Q. Give reasons for the following.

- (A) Danger signals installed at airports and at the top of tall buildings are of red colour.
- (B) The sky appears dark to the passengers flying at very high altitudes.
- (C) The sun appears reddish during sunrise and sunset.
- (D) School buses are painted yellow in colour.

Ans. (A) Danger signals are red because red light is least scattered by air due to its longer wavelength, making it visible from far distances, even in fog or haze.

(B) The sky appears dark at high altitudes because the atmosphere is very thin, so there are very few air molecules to scatter sunlight.

(C) The Sun appears reddish during sunrise and sunset because sunlight has to travel a longer distance through the atmosphere, and blue light is scattered away, leaving mostly red and orange light to reach our eyes.

(D) School buses are painted yellow because yellow has a high visibility and is easily noticed even in fog or dim light due to its wavelength being next to red, ensuring better safety.

**"EVEN YOUR BLIND SPOT
HAS A JOB—REMINDING YOU
THAT NO ONE IS PERFECT,
NOT EVEN YOUR AMAZING
EYES."**

- embrace your flaws

